



OPEN A lightweight convolutional neural network for real-time monitoring of smart mango orchard systems

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Recent advancements in Convolutional Neural Networks (CNNs), combined with the growing adoption of farm-applicable Internet of Things (IoT) devices, have expanded the application of precision agriculture in mango orchards. The Smart Mango Orchard can play a crucial role in ensuring mango trees thrive and produce high-quality fruit. However, state-of-the-art (SOTA) CNNs are built on numerous layers and many parameters; therefore, they are challenging to deploy in IoT devices. However, the lightweight CNN is a possible solution. This study developed a lightweight CNN, mangoNet, to deploy in an innovative mango orchard environment. The mangoNet is expected to monitor the mango leaf images and report them to farmers via a mobile app using the IoT system. The study was conducted using the primary dataset collected from the mango gardens in Rajshahi, Bangladesh. The mangoNet benchmark was evaluated using six SOTA CNNs. The mangoNet, with only 3,987,400 Parameters, outperforms SOTA CNNs' accuracy (99.61%). In addition, this study employed SHAP, LIME, and Grad-CAM visualizations to identify and depict the image regions that contribute to mangoNet's decision-making process. The mangoNet is integrated into a Streamlit web application and an Android mobile app, as researchers suggest for the practical use of CNNs. The novelty of mangoNet lies in its balanced sequential architecture, with a careful selection of kernels and progressive filter expansion, enabling early layers to capture low-level features and deeper layers to extract high-level features. As a result, the proposed mangoNet achieved high accuracy while requiring fewer computational resources and reduced training time. In addition, the mangoNet-powered website and mobile application empower both farmers and farming stakeholders by making real-time disease detection. In the future, the prototype is expected to be commercialized as Bangladesh is the 8th mango-producing country in the world.

Keywords Mango leaf detection, Precision agriculture, Convolutional neural network, CNN, Lightweight CNN, SHAP, LIME, Grad-CAM

Bangladesh is the world's 7th-largest mango-growing country. Mango orchards are the primary source of income in the mango-growing zone, as other crops are not suitable for that land and weather. Mangoes are susceptible to leaf diseases, which significantly impact their quality and production. Mango leaf disease slowly spreads into the tree and the orchard. Diagnosing mango leaf diseases with the human eye can be difficult, time-consuming, and often inaccurate. The mango disease is caused by variations in environmental parameters¹ and by climate change². However, the frequency of mango disease is increasing, and its intensity is very severe. Early prediction of disease on mango leaves can help farmers prevent potential losses.

Convolutional Neural Networks (CNNs) have revolutionized precision agriculture by enabling accurate and efficient disease identification^{3,4}. Among deep CNNs, two forms of CNN, the ensemble model, which combines multiple models, and the transfer learning method, are highlighted as effective methods in mango leaf disease detection. However⁵, criticized deep CNN architectures for having a large memory footprint and high computational demand, a criticism also supported by⁶. Lightweight CNNs (L-CNNs), therefore, aim to reduce computational complexity and memory usage⁷. L-Nets are mainly developed using (1) network architecture^{5,8,9}, (2) nonlinear activation functions^{7,10}, (3) supervision components¹¹, (4) regularization mechanisms¹², and

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(5) optimization techniques¹³. However, L-Net has recently attracted significant attention from researchers in mango leaf detection and classification, e.g.^{14–17}, who have presented L-Net for mango leaf classification.

With the proliferation of the Internet of Things (IoT) and the adaptability of CNNs on edge devices in IoT settings, researchers are also proposing a smart mango orchard for real-time mango disease monitoring and management. The combined technologies of the Internet of Things (IoT) and CNNs can offer scalable, sustainable strategies for disease detection and management. Such technology is expected to bridge the gap between traditional and smart mango orchard management. High-speed internet, advancements in sensors, applications such as radio frequency identification (RFID) and wireless sensor networks (WSNs), and improved mobile technologies, combined, present an opportunity to transform traditional mango orchard management into a smart mango orchard. The advantages of a smart mango orchard can be achieved via deep learning. In such a system, computations are performed on an edge device, which reduces power consumption and latency. A smart mango orchard integrated with deep learning has the potential to create an enabling environment for early detection and real-time monitoring of mango leaf disease.

Another extension of CNN is the integration of explainable artificial intelligence (XAI) to visualize the decision-making process of CNN models, thereby developing transparency and interpretability¹⁸. XAI visualization illustrates how CNN aids in understanding the model's focus areas during decision-making; these techniques aim to identify the regions and features in input images that contribute most to the model's predictions¹⁹. Local interpretable model-agnostic explanations (LIME), Shapley additive explanations (SHAP), and Grad-CAM are some XAI.

The core problem of this study is that many studies have attempted to develop lightweight CNNs; a standard benchmark for lightweight CNNs is accuracy. For example¹⁴, developed a lightweight CNN with 101,000 parameters that achieves a 99.14% success rate in disease leaf identification. However, edge devices require efficient processing, reduced storage, and limited computational requirements from the mobile application, which can be achieved through edge computing without using the cloud²⁰. However, these analyses are often not presented in a lightweight CNN. Furthermore, reducing ML models' size without significant loss in accuracy is another important benchmark for ML Models on edge devices. In addition, the performance evaluation used images captured with high-resolution cameras or in a controlled environment, which are far from the real conditions that edge devices might encounter. Inference time after the model is loaded from disk into memory is also a critical aspect²¹. Another limitation of existing lightweight CNNs is that although many studies produce high-accuracy models, they do not report performance on unseen real-world mango leaf diseases. The gap exists in lightweight CNNs that balance model accuracy with resource efficiency.

Several motivating factors are involved in this study. The gradient losses in the CNN's later layers motivated us to investigate a balanced CNN architecture with kernel and filter expansion, enabling early layers to capture low-level features and deeper layers to extract high-level features. We aimed to integrate the MaxPooling and dropout layers to provide robust feature learning and regularization, stabilize gradient flow, and reduce the risk of vanishing gradients during CNN training. The complex, intricate network architectures of CNNs are a limitation for applying them on edge and mobile devices. This leads the researchers to use a lightweight CNN.

Recent advancements in CNNs, combined with the growing adoption of farm-applicable Internet of Things (IoT) devices, have expanded the application of precision agriculture in mango orchards. The Smart Mango Orchard can play a crucial role in ensuring mango trees thrive and produce high-quality fruit. However, state-of-the-art (SOTA) CNNs are built on numerous layers and many parameters; therefore, they are challenging to deploy in IoT devices. However, the lightweight CNN is a possible solution. This study developed a lightweight CNN, mangoNet, to deploy in an innovative mango orchard environment. The mangoNet is expected to monitor the mango leaf images and report them to farmers via a mobile app using the IoT system. The other challenges are as follows:

1. Mango leaf disease detection and classification face significant challenges due to the small and irregularly shaped disease-affected regions and variations in lesion patterns across different diseased classes. Previous studies by^{22,23} also identified challenges for CNN models, in which location-specific and spatial details are lost in deeper layers.
2. Accuracy in mango leaf detection is a concern, as lower detection accuracy and large false-positive values will narrow the CNN's applicability and acceptability in mango leaf research²⁴. Earlier²⁵, also cautioned about the low accuracy and high false-positive rates in mango leaf disease detection.
3. Some research is conducted on small datasets and limited mango leaf disease modalities. For example²⁶, analyzed a dataset of only 1,100 mango leaf images, whereas²⁷ used 2,918 photos, limiting the generalizability of their models. Similarly¹⁷, focused solely on specific mango leaf diseases, and²⁸ considered only two disease modalities.
4. SOTA CNNs' performance in detecting mango leaf diseases is still unclear, as they were tested on large-scale datasets such as PlantVillage or PlantDoc. The images of these datasets differ significantly from those of mango leaves collected from a given mango orchard. The study by¹⁸ emphasized evaluating existing SOTA CNNs for plant disease detection in specialized domains.
5. The authors of the study²⁹ noted that in mango leaf detection using CNN models, the diseased regions need to be visualized to understand the decision-making process of the CNN models. CNN models extract features for classification by leveraging color, pixel intensity, and pixel location. The authors of the study³⁰ argued that the insufficient explainability makes it challenging to interpret the models' outputs.

Following these gaps, this study conducts the following experiments, which make the following contributions and novelties of this study:

1. This study introduces mangoNet (Minimal Net), a lightweight CNN model with fewer layers and learnable parameters that can reliably detect mango leaf diseases quickly.
2. Along with the mangoNet, six SOTA CNNs from spatial exploitation (VGG19), depth-based (ResNet152v2), multi-path (DenseNet121), width-based multi-connection (ResNext101), and Feature-Map Exploitation (SE-ResNet152) were applied to the Mango Leaf Disease dataset to investigate the most effective learning architecture from images.
3. LIME, SHAP, and Grad-CAM were applied to generate explanations for the CNN detection and classification.
4. Earlier, the authors of the study³¹ advocated that pixel intensity has an impact on CNN performance, and to establish the scientific result, this study analyzes the correctly classified and misclassified groups, true positive, true-negative, false positive, and false negative groups.

The rest of the paper is organized as follows. Section 2 provides a literature review; Sect. 3 outlines the details of the experiments; Sect. 4 presents the results of the experiments; Sect. 5 is the discussion; Sect. 6 is the limitations and future research direction; and Sect. 7 is the conclusion.

Related works

The literature review of this study followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. A review of around thirty-five selected papers is presented in this paper. All of the articles cover the classification and detection of plant leaves using transformer-based deep learning techniques. Figure 1 and 2 describe the process of article selection:

Articles were selected for final review using a three-stage screening process based on a series of inclusion and exclusion criteria. After removing duplicate records generated by using two databases, articles were first screened solely by title. The abstract was assessed, and the full articles were then checked to confirm eligibility. The chief investigator conducted the entire screening process.

Included articles were limited to those published from 2018 to 2025 to focus on deep learning methodologies. Here, a study was defined as work that employed a transformer-based deep learning algorithm to classify or detect plant leaf disease and used one or more of the following performance metrics: accuracy, the area under the receiver operating characteristic curve, sensitivity, specificity, or F1 score.

Exclusion criteria were: preprint studies

A group of studies compared CNN models for leaf disease classification. For example³², evaluated the performance of several CNN models for apple leaves classification. The study reports that the accuracy values obtained show that all the discussed models are very accurate. However, the EfficientNetV2_m model is the best, achieving 100% accuracy and an F1 score of 100%. These studies reveal that, compared with conventional machine learning methodologies, the outcomes of deep learning methods can be much more beneficial for the agricultural sector, providing a vision for how they can be incorporated and used to improve various farm management functions.

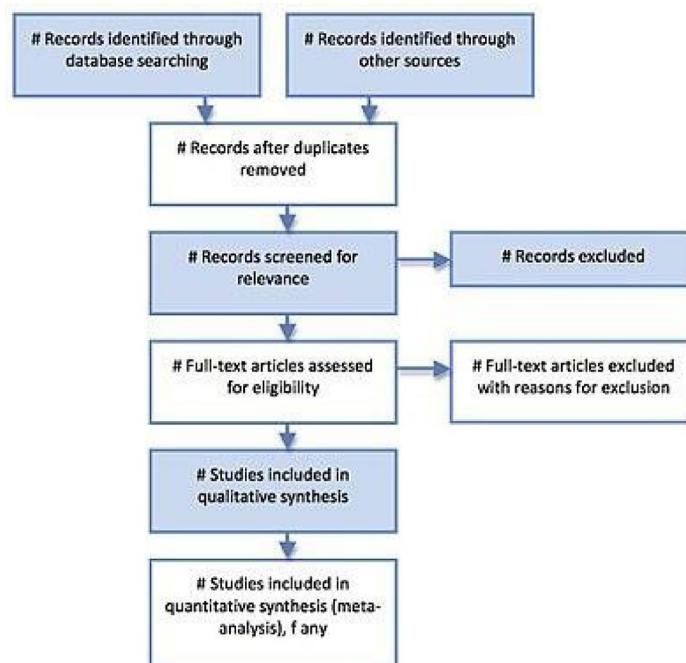


Fig. 1. The process of article selection.

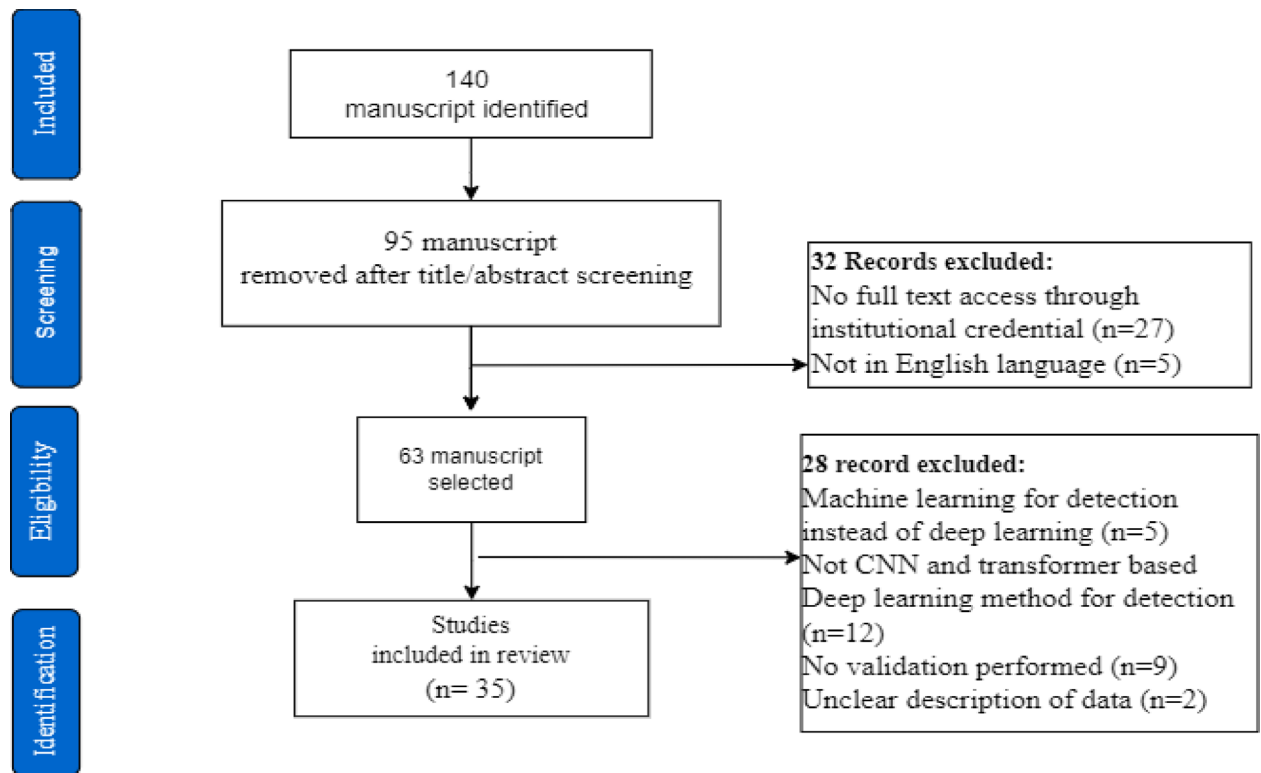


Fig. 2. The Preprint studies.

Earlier, Ahad et al. (2023) also compared the performance of SOTA CNNs for rice leaf disease detection. In another study³², used ResNet architectures to identify tomato diseases. The study focused on the capabilities of models such as Res2Next50 and ResNet50d. The results of the study reveal that the Res2Next50 model achieved the highest accuracy of 99.85% and an impressive F1 score of 99.82%. This indicates a high level of precision and robustness in distinguishing between healthy and diseased leaves. Additionally, the Res2Net50d model has proven highly effective, achieving 99.78% accuracy, 99.73% precision, and 99.72% recall.

Among CNN applications for mango image detection and classification²⁶, explored the use of CNN models to detect 13 distinct fungal and bacterial infections in mango leaves. The dataset, comprising 1,100 images, was collected from the Chittoor district of Andhra Pradesh, one of India's largest mango-growing regions. The study employed GoogLeNet, EfficientNet, and ResNet-50, among which EfficientNet achieved the highest accuracy of 98.7%. Rizvee et al. (2023) developed a CNN-based model, LeafNet, to detect seven prevalent mango leaf diseases using a dataset specific to Bangladesh's mango-growing regions. LeafNet achieved an average accuracy of 98.55%, precision of 99.50%, recall of 99.45%, F1-score of 99.47%, and specificity of 99.88%. The study³³ introduced a multi-scale and multi-pooling CNN (MSMP-CNN) model to identify mango leaf diseases. The proposed model achieved 95% accuracy, which increased to 98.5% after applying transfer learning.

Ensemble and hybrid models are also applied in mango leaf disease detection and classification. For example³⁴, developed an ensemble model that integrates GoogLeNet and VGG16. The model achieved 99.87% training accuracy, 99.72% validation accuracy, and 99.21% test accuracy. The study³⁵ proposed an overlapped patch embedding approach to address the discontinuity issue in Vision Transformers (ViTs) caused by the use of non-overlapping patches³⁶. ensemble ViT and DeiT models and achieved an accuracy of 99.75%. This indicates the potential to achieve optimal coverage with fewer epochs, thereby reducing training time. The study by³⁷ compared the performance of CNNs and Vision Transformers (ViTs) and utilized transfer learning, fine-tuning, and data augmentation. The best-performing model, EfficientNetV2S, achieved 91% accuracy, an F1-score of 0.92, and an Intersection over Union (IoU) of 0.83 for disease localization. The study by²⁹ introduced WaveVisionNet, a multi-layer perceptron model. The model was evaluated on MangoFolioBD and publicly available datasets, achieving an accuracy of 96.11%.

Feature extraction-based methods for mango leaf disease have also received significant attention. For example, the authors of the study³⁸ developed FrCNN, a fully CNN (FCN)-based approach for segmenting diseased regions in mango leaves. FrCNN learns pixel-level features directly from input images. The model was compared with VGG16, VGG19, and U-Net. The model achieved an accuracy of 99.2%. The study³⁰ introduced a hybrid deep learning model, XAI-FruitNet, that improves fruit classification by combining average and max pooling. The model was evaluated on four datasets, including Fruits-360 and Fruit Recognition, achieving an accuracy of over 97% and outperforming existing models. The study³⁹ diagnosed mango leaf diseases using Total Variation Filter-based Variational Mode Decomposition (TVF-VMD). The TVF-VMD method extracts disease-related features using DenseNet121 and VGG-19, then combines them and classifies them using the Random

Forest algorithm. The model achieved 98.85% accuracy. The study⁴⁰ addressed automatic plant leaf disease detection using a Deep CNN to improve feature representation and correlation, and a Generative Adversarial Network (GAN) for data augmentation to address data imbalance. Based on accuracy, precision, recall, and F1 score, which have significantly improved over conventional methods for the plant leaf disease database (accuracy of 99.74%, precision, recall, and F1 score of -0.99), the success of the suggested strategy is assessed.

The field of lightweight CNNs for agriculture has advanced significantly in recent years. Research has focused on reducing the parameter count of CNNs while maintaining high classification accuracy. Models like ResNet50, DenseNet121, and VGG16 have been adapted for plant disease classification, but their high computational demands make them unsuitable for real-time applications in IoT environments. Lightweight models such as mangoNet, MobileNetV2, and EfficientNet offer promising solutions by minimizing parameter count without compromising accuracy. Recent studies such as^{9,23} have demonstrated the effectiveness of lightweight CNNs for crop disease detection, focusing on improving the efficiency of CNN models for field applications. These studies highlight the growing need for models that can operate efficiently on edge devices with limited computational resources, a key gap that mangoNet addresses. The study⁴¹ presented a new, deeper, and lightweight convolutional neural network architecture (DLMC-Net) for plant leaf disease detection across multiple crops in real-time agricultural applications. The passage layer and a series of collective blocks are added to the suggested model to extract deep-level characteristics. These advantages include feature reuse and propagation, which help address the vanishing gradient problem. Point-wise, separable convolution blocks are also used to reduce the number of trainable parameters. On eight metrics, including accuracy, error, precision, recall, sensitivity, specificity, F1-score, and Matthews correlation coefficient, experimental results of the proposed model are compared against seven state-of-the-art models. Even with complex background conditions, the suggested model outperformed all others, achieving accuracies of 93.56%, 92.34%, 99.50%, and 96.56% on datasets for citrus, cucumber, grapes, and tomatoes, respectively. The study by⁴² proposed a lightweight CNN (LCNN) to detect mango leaf disease. The authors classified seven mango leaf diseases and healthy leaves using pre-trained models VGG16, Resnet50, Resnet101, and Xception. The LCNN model achieved the highest accuracy, 98%, which was better than that of other pre-trained models. The study by²⁷ proposed a new method for identifying and categorizing mango leaf illnesses using a CNN trained via federated learning. The suggested model for four distinct customers, focusing on the five disease classifications: Healthy, Anthracnose, Powdery Mildew, Leaf Spot, and Leaf Curl. The experiment results show Precision values ranging from 93.33% to 96.01%, recall values ranging from 90.59% to 97.45%, F1-scores ranging from 92.64% to 96.10%, and accuracy values ranging from 97% to 98%. The macro, weighted, and micro averages range from 93.18% to 94.97%, 93.26% to 95.08%, and 93.26% to 95.08%, respectively. By employing federated learning to protect data privacy, this methodology enables customers to collaborate and benefit from shared learning without risking their data.

The study²⁸ analyzed the effects of transformer-based approaches that aggregate attention at different scales on feature variants for the classification of tomato leaf diseases from image data. Four state-of-the-art transformer-based models, namely, External Attention Transformer (EAnet), Multi-Axis Vision Transformer (MaxViT), Compact Convolutional Transformers (CCT), and Pyramid Vision Transformer (PVT), are trained and tested on a multiclass tomato disease dataset. The results analysis shows that MaxViT comfortably outperforms the other three transformer models, achieving 97% overall accuracy, compared with 89% for EAnet, 91% for CCT, and 93% for PVT. The MaxViT architecture is the most effective transformer model for classifying tomato leaf disease because it exhibits a smoother learning curve than other transformers.

The study¹⁸ developed a novel model, the Vision Transformer (ViT), replacing a CNN for classifying cassava leaf diseases. On the cassava leaf disease dataset, experimental results demonstrate that this model achieves competitive accuracy, at least 1% higher than well-known CNN models such as EfficientNet and Resnet50d. This research also showed that the ViT model can be successfully deployed on the Raspberry Pi 4, an edge device that can be connected to a drone, enabling farmers to quickly and effectively identify diseased leaves.

The study²⁹ proposed an automatic pest identification method based on the Vision Transformer (ViT). The plant disease and insect pest data sets are improved using techniques including Histogram Equalization, Laplacian, Gamma Transformation, CLAHE, Retinex-SSR, and Retinex-MSR to prevent overfitting during training. According to the simulation results, the built-in ViT network achieves a test recognition accuracy of 96.71% on the publicly available Plant_Village dataset of plant diseases and insect pests, which is about 1.00% higher than that of a method based on conventional CNNs such as GoogleNet and EfficientNetV2.

The study³⁰ proposed a residual-distilled transformer architecture for feature extraction and prediction. A multi-layer perceptron (MLP) is fed with the residual concatenation of the vision transformer and the distillation transformer. On the rice leaf disease dataset collected in paddy fields, the proposed method outperforms the current state-of-the-art models, achieving an F1 score of 0.89 and top-1 accuracy of 0.92.

The study³¹ presented a lightweight, Shuffle-convolution-based vision transformer for effective diagnosis of sugarcane leaf diseases, named SLViT. The SLViT hybrid network is initially trained on the freely available PlantVillage disease dataset and the independently created SLD10k sugarcane leaf disease dataset. The transformer encoder is converted to a flexible plug-in (LViT) and then integrated into several locations of a lightweight CNN architecture (SHDC). The experiments show that all of SLViT's modifications have improved the system's overall performance. On Plant Village, SLViT outperforms three specially created leaf-disease recognition models and six SOTA models in terms of speed (1,832 FPS), model size (2 MB), power consumption (50 mW), and precision (98.84%). On the SLD10k dataset, SLViT outperformed MobileNetV3_small by 1.87% in accuracy and reduced model size by 66.3%.

The study²³ developed a Vision Transformer-enabled CNN model called "PlantXViT" for plant disease identification. The suggested model effectively recognizes a wide range of plant diseases across various crops by combining the strengths of conventional CNNs with Vision Transformers. The suggested model is appropriate for IoT-based innovative agricultural services, as it has a lightweight structure and only 0.8 million trainable

parameters. On all five datasets, the proposed PlantXViT network outperforms five cutting-edge techniques. Even under difficult background conditions, the average accuracy for identifying plant diseases exceeds 93.55%, 92.59%, and 98.33% on datasets for apples, maize, and rice, respectively. Gradient-weighted class activation maps and Local Interpretable Model-Agnostic Explanations are used to assess the model's explainability. The authors of the study³ focused on fine-tuning cutting-edge pre-trained CNN and vision transformer models to classify the PlantVillage dataset, which comprises 4062 leaf images distributed across four categories of grape leaves, and to diagnose. Moreover, the study used the Grapevine dataset, consisting of 500 leaf images categorised into five grape leaf types, with 100 images per type. In experiments with 14 CNN and 17 vision transformer models, deep learning demonstrated high accuracy in distinguishing grape diseases and recognising leaves. Notably, four models achieved 100% accuracy on PlantVillage and Grapevine datasets, with SwinV2-Base standing out. This approach holds promise for enhancing crop productivity through early disease detection and providing insights into grape variety characterization in agriculture. Earlier³², also combined CNN and ViT to identify and categorize olive leaves. The study reports 96% accuracy in multiclass classification and 97% in binary classification; the model outperformed the competition. The authors of the study³³ advocated implementing IoT in agriculture to enable automation and real-time farm monitoring. The study proposed an IoT framework for the apple orchard to enable comprehensive automation using advanced IoT technologies and robotics. The reason for the automation is unexpected weather conditions, including rain, hailstorms, droughts, and fog, which often cause apple leaf infections, leading to a significant decrease in fruit production³⁴. The IoT-based automated orchard will increase economic benefits, improve overall quality, and boost production, thereby generating economic benefits for a country.

Experimental method and materials

In this study, we aim to develop a lightweight CNN mangoNet to implement in the mango orchard for Smart IoT-based and mangoNet-based precision agriculture. Therefore, two essential criteria for mangoNet are: first, the model must be lightweight and edge-device-integratable, and second, it must achieve high accuracy on unseen datasets. The mangoNet model was tested on a primary mango leaf disease dataset and a publicly available dataset. The mangoNet model performance was compared with six (6) prominent CNNs, transfer learning, and two ensemble models. We also conducted computational cost and statistical analysis of mangoNet with SOTA CNNs to confirm its robustness. The study's workflow is presented in Fig. 3. We first describe the hardware and software for the study.

The following sections describe the hardware specification, dataset description, mangoNet model development, and training procedure.

Hardware and software specifications

The experiments were conducted on a Dell Precision 7680 mobile workstation equipped with a 13th-generation Intel Core i9-13950HX vPro processor running Windows 11 Pro, an NVIDIA RTX 3500 Ada Generation graphics card, 32 GB of DDR5 memory, and a 1 TB SSD. Python (version 3.9) was used as the programming language, as this version supported TensorFlow-GPU, SHAP, and LIME.

Dataset description

In this study, two sets of mango leaf disease datasets were used. The primary dataset (Dataset A) was collected from mango orchards in Kushtia and Dhaka, Bangladesh, two major mango-growing regions in the country. Data collection occurred between August 19 and August 22, 2023. The corresponding and principal author is the owner of this dataset, which will be made available upon request. Additionally, the study utilized publicly available data from the following <https://www.kaggle.com/datasets/aryashah2k/mango-leaf-disease-dataset>.

Tree selection was based on accessibility, leaf availability, and maturity, ensuring a sufficient representation of the targeted diseases. While trees were randomly chosen, priority was given to those with an adequate number of affected leaves. Only mature and visibly diseased leaves were included in the dataset. After collection, leaves were sorted into disease-specific classes under expert guidance from the Bangladesh Agriculture Association to ensure accurate classification. Eight labelled buckets were used for sorting, and care was taken to select undamaged, non-rotten leaves. A dataset of healthy and diseased mango leaves was compiled, covering diseases such as Anthracnose, Bacterial Canker, Cutting Weevil, Die Back, Gall Midge, Powdery Mildew, and Sooty Mould.

The leaves were placed on a white background for high-resolution photography. Special attention was given to image clarity, as it is crucial to reliable identification and classification. The process classification of diseased photos is described in Fig. 4. Sample images are shown in Fig. 5, and the distributions of the training, validation, and test sets are shown in Fig. 6.

The second dataset, Dataset B, is a public dataset on mango leaf disease available on Kaggle (<https://www.kaggle.com/datasets/aryashah2k/mango-leaf-disease-dataset>). The dataset also had eight (8) classes and 500 images in each class. The class is similar to our primary data, as the dataset was also collected in a Bangladeshi mango orchard. The photos are of 240 × 320 mango leaves in JPG format. A total of seven classes of disease are included: Anthracnose, Bacterial Canker, Cutting Weevil, Die Back, Gall Midge, Powdery Mildew, and Sooty Mould. The mango leaf images were collected from the orchards of Sher-e-Bangla Agricultural University, Jahangir Nagar University, Udaypur village, and Itakhola village in Bangladesh.

Image preprocessing

To enhance the visibility of imaging features and reduce noise, a multistep preprocessing pipeline was applied. Contrast Limited Adaptive Histogram Equalization (CLAHE) was employed to improve local contrast in dense tissue regions, followed by Gaussian blurring to suppress high-frequency noise, defined as:

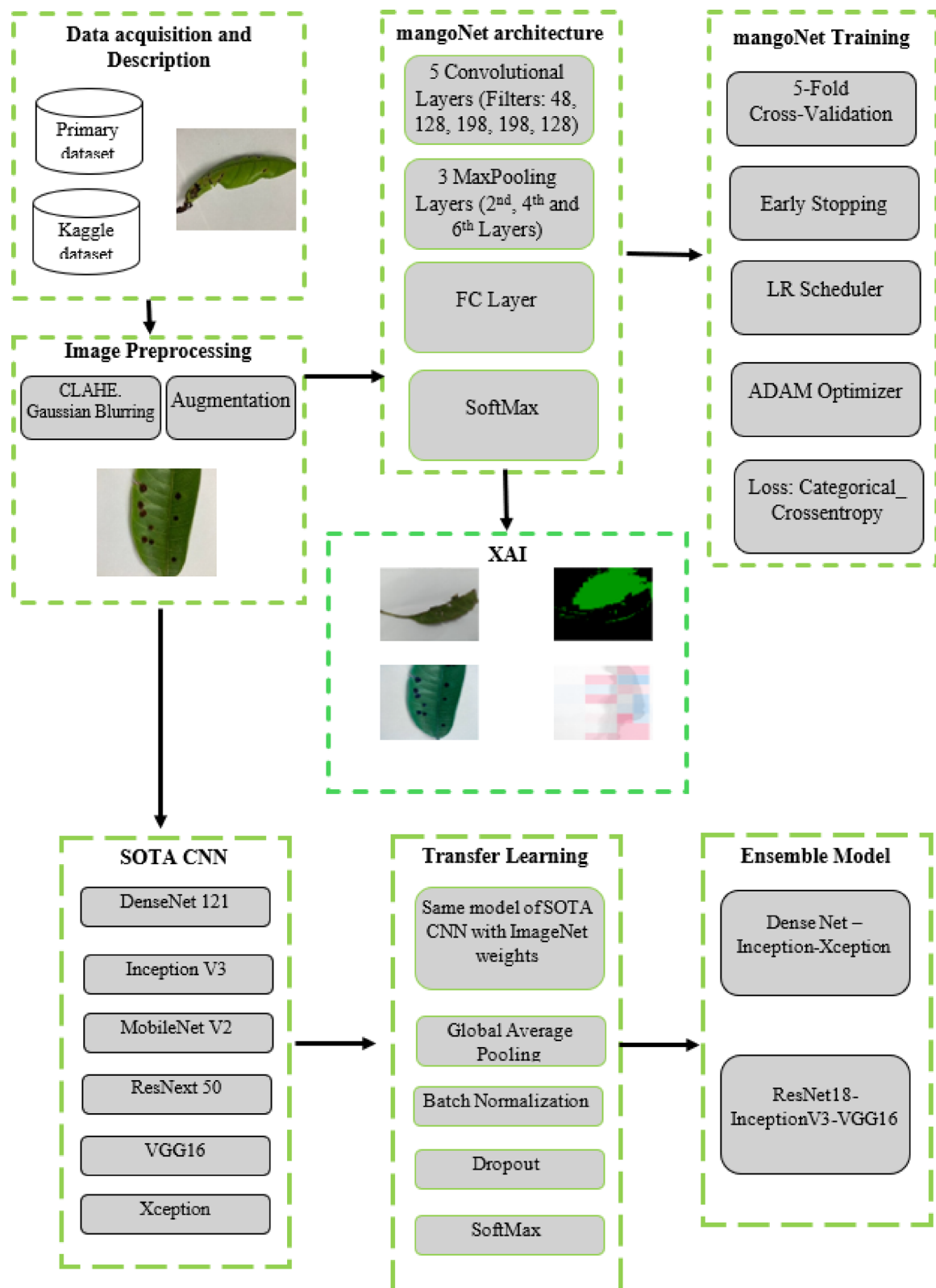


Fig. 3. Workflow of the proposed methodology.

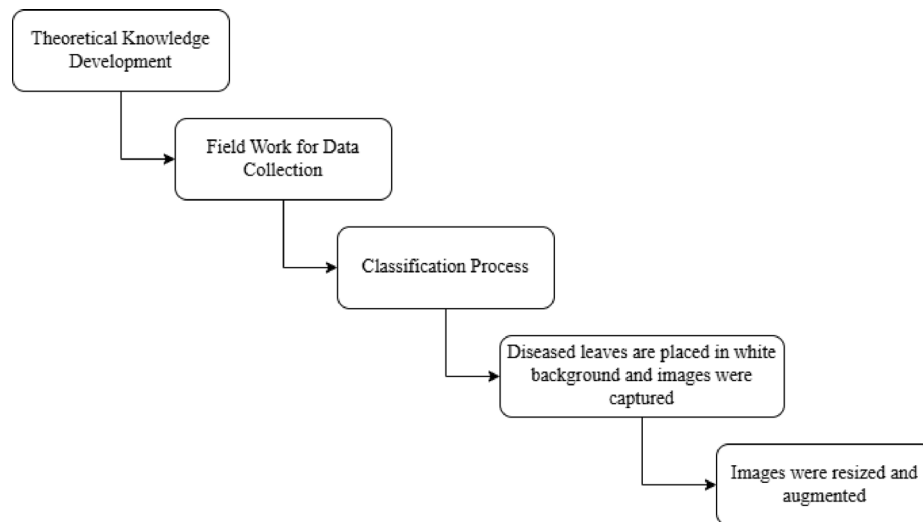


Fig. 4. Process of dataset preparation.

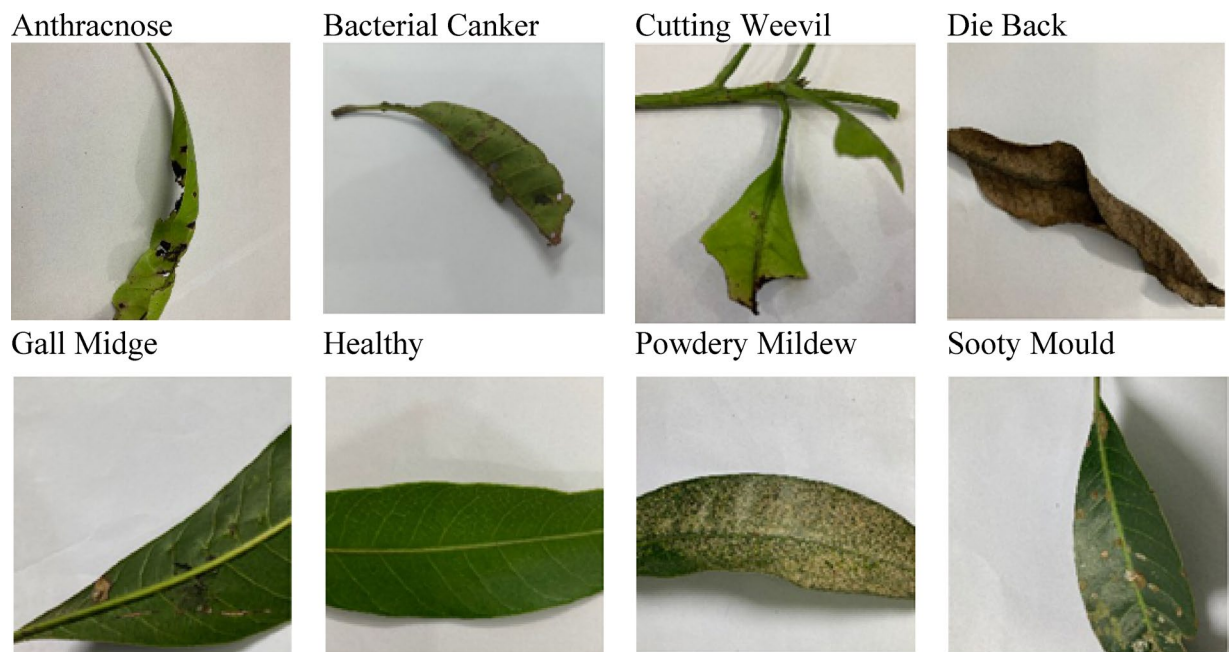


Fig. 5. Examples of 8 mango leaf classes.

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (1)$$

Where σ represents the standard deviation of the Gaussian distribution, subsequently, bilateral filtering was applied to preserve edge structures critical for mass detection, expressed as:

$$I_{filtered}(x) = \frac{1}{Wp} \sum_{xi \in \Omega} I(xi) \cdot fr(|I(xi) - I(x)|) \cdot fs(|xi - x|) \quad (2)$$

Where fr denotes the range kernel, fs the spatial kernel, and Wp the normalization factor. Non-local means denoising was used to enhance smoothness further while preserving important features. Finally, unsharp masking was performed to improve edge clarity, formulated as:

$$I_{sharp} = I_{original} + \alpha \cdot (I_{original} - G_{blur})$$

Here, $\alpha = 1.5$ controls sharpness. (3)

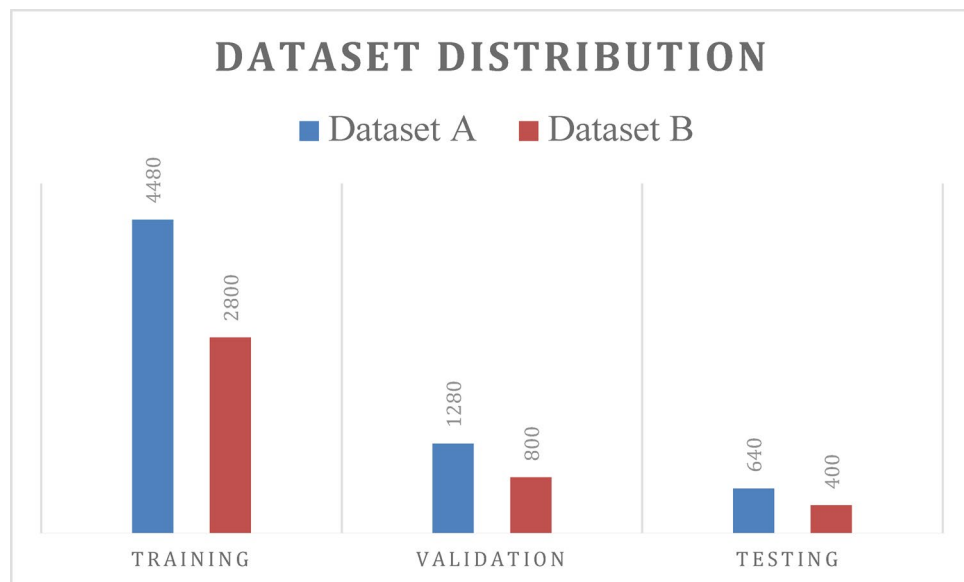


Fig. 6. Distribution of images.

To mitigate overfitting and enhance generalization, a unified set of augmentation techniques was applied across all datasets, including geometric transformations ($\pm 15^\circ$ rotations, center cropping) and photometric adjustments (brightness/contrast modulation, random scaling). Horizontal and vertical flipping were incorporated to increase dataset diversity, as commonly practiced in mammography deep learning to simulate orientation variations^{39,40}. To address concerns regarding potential anatomical distortion from flipping (e.g., altering medial-lateral breast positioning), a controlled experiment was conducted by retraining MangoNet without horizontal flips, retaining clinically aligned augmentations such as $\pm 15^\circ$ rotations (mimicking patient positioning variability) and CLAHE-based contrast adjustments (reflecting equipment differences).

Image augmentation

Image augmentation has been a practical approach to enhancing the robustness of CNN models. This study uses image transformations, such as scaling, rotation, and contrast adjustment, to enhance training datasets and improve the performance of CNNs across a range of complex tasks. Specifically, in the context of mango leaf disease detection, the technique allows models to handle better variability in leaf disease presentation, including differences in size, shape, and location^{22,23,25–27,35}.

Design choice and developing mangoNet for an IoT environment

Since the aim of the study was to develop a lightweight CNN for an IoT environment, ensuring efficient operation on resource-constrained IoT devices was the primary goal. Firstly, the model was created using a sequential CNN. A sequential design simplifies the network and reduces computational complexity. mangoNet CNN is designed to extract low-level to high-level features of the mango leaves. It first focuses on the edges, then on textures, and finally on objects or spots on the leaves. Each layer feeds into the next, providing a linear computational flow. Moreover, the sequential structure ensures faster training and inference times. mangoNet includes five convolutional layers with progressively expanding filter sizes (48, 128, 198, 128) (see Fig. 7 for a layered view and Fig. 8 for visualization). This layered approach helps the model effectively capture low-level features (edges, veins) in the early layers and high-level features (disease spots) in deeper layers. Number of kernels used: 48, 128, 198, and 128, and size 3×3 . The 3×3 kernel size captures slightly smaller spatial features than 4×4 (with 9 parameters per filter compared to 16). This balance improves the model's performance while keeping the number of parameters low. MaxPooling layers were used to achieve spatial invariance, ensuring that the model remains robust to changes in leaf orientation or size. The use of Dropout (50%) and early stopping ensures the model does not overfit. The Adam optimizer was chosen for its ability to converge quickly and effectively on datasets of limited size, ensuring optimal performance even in resource-limited settings. These choices are supported by empirical evidence in the literature, such as²⁸, who demonstrated that lightweight CNNs optimized with Adam perform well in agricultural disease detection.

After preprocessing, the mango leaf images were resized to 224×224 , and the input images were fed to the first Conv2D layer (with 48 filters) to detect basic textures, such as edges, veins, and spots. The mangoNet image X is fed into the layer and processed by the 1st convolution layer:

$$\text{Conv1}(X) = \text{ReLU}(\text{Conv2D}(X, W_1) + b_1)$$

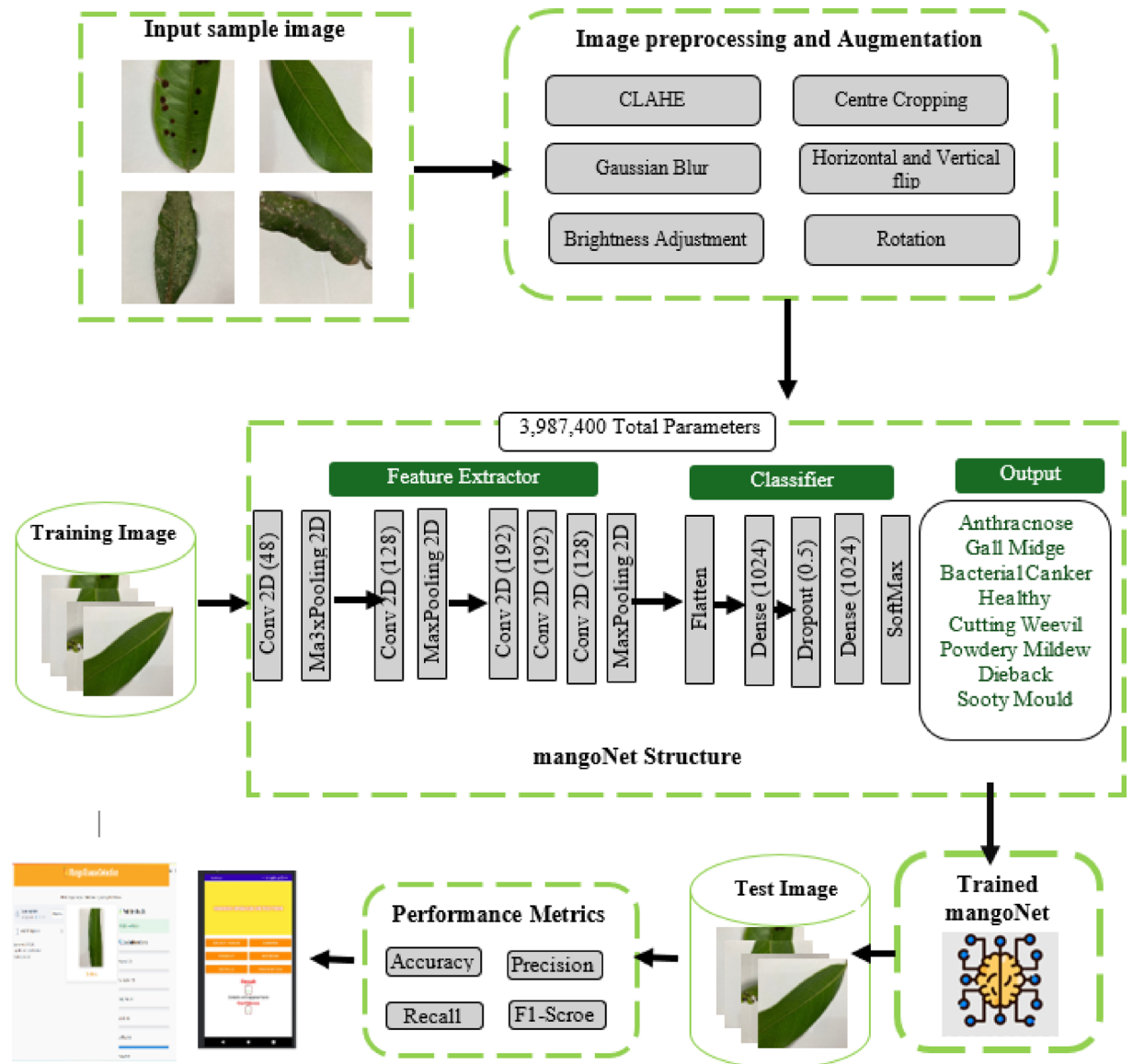


Fig. 7. Layered architecture of mangoNet.

After the convolutional layer, a MaxPooling2D layer is applied to provide invariance to leaf orientation and size. MaxPooling progressively reduces feature maps from $110 \rightarrow 36 \rightarrow 12 \rightarrow 4$. The Max Pooling Layers equation can be expressed using the following equation:

$$\text{Pool1}(X) = \text{MaxPool}(\text{Conv1}(X), 3 \times 3)$$

The pooled feature map then serves as the input to the second convolutional layer, allowing Conv2 to capture more complex shapes, like the outline of a disease area, the curve of tissue structures, or other meaningful regions in the image detected by Conv1:

$$\text{Conv2}(X) = \text{ReLU}(\text{Conv2D}(\text{Pool1}(X), W_2) + b_2)$$

$$\text{Pool2}(X) = \text{MaxPool}(\text{Conv2}(X), 3 \times 3)$$

These feature maps are then fed into the 5th, 6th, and 7th layer progressively to detect higher to lower-level patterns (shapes, parts of objects, object-specific details). In these layers, higher-level structures and more complex spatial relationships are learned:

$$\text{Conv3}(X) = \text{ReLU}(\text{Conv2D}(\text{Pool2}(X), W_3) + b_3)$$

$$\text{Conv4}(X) = \text{ReLU}(\text{Conv2D}(\text{Pool3}(X), W_4) + b_4)$$

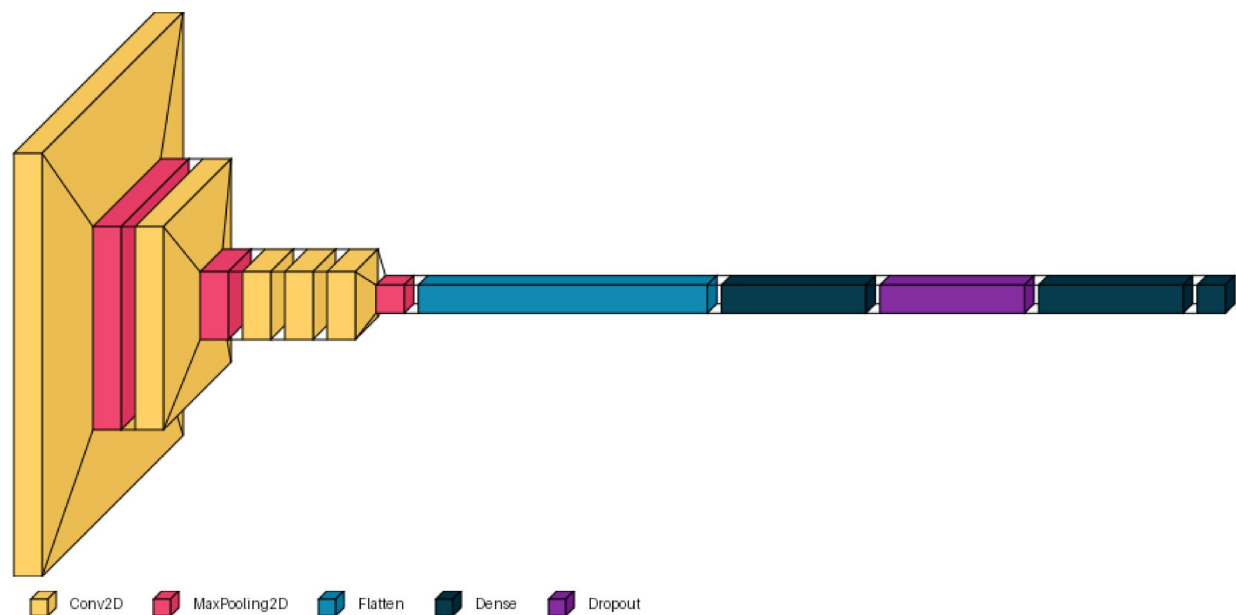


Fig. 8. mangoNet model visualisation.

Layer Type	Output Shape	Param #
Conv2D (conv2d_5)	(110, 110, 48)	1,344
MaxPooling2D (max_pooling2d_3)	(36, 36, 48)	0
Conv2D (conv2d_6)	(36, 36, 128)	55,424
MaxPooling2D (max_pooling2d_4)	(12, 12, 128)	0
Conv2D (conv2d_7)	(12, 12, 192)	221,376
Conv2D (conv2d_8)	(12, 12, 192)	331,968
Conv2D (conv2d_9)	(12, 12, 128)	221,312
MaxPooling2D (max_pooling2d_5)	(4, 4, 128)	0
Flatten (flatten_1)	(2048)	0
Total Params		3,987,400
Trainable Params		3,987,400
Non-trainable Params		0

Table 1. Presents the architecture details of the proposed mangoNet model. Summary of each layer, including type, output shape, and parameter count. Architecture details of the mangoNet model

$$\begin{aligned} \text{Conv5}(X) &= \text{ReLU}(\text{Conv2D}(\text{Pool2}(X), W_5) + b_5) \\ \text{Pool3}(X) &= \text{MaxPool}(\text{Conv3}(X), 3 \times 3) \end{aligned}$$

The output of Pool3 is then passed to the Flatten Layer, which converts 2D feature maps into a 1D vector. The feature vectors are required to feed into dense layers. The Equation is expressed as:

$$X_{\text{flat}} = \text{Flatten}(\text{Conv4}(X))$$

This flattened output feeds into a dense layer with 1024 neurons, which learns complex feature interactions and higher-level representations:

$$X_{\text{dense}} = \text{ReLU}(X_{\text{flat}} \cdot W_d + b_d)$$

A dropout layer with a rate of 0.5 follows, meaning that 50% of the neurons are randomly set to zero during training. This is mathematically represented as each neuron having a probability $p = 0.5$ of being dropped effectively reduces the risk of overfitting by forcing the network to learn more robust representations.

$$X_{\text{drop}} = \text{Dropout}(X_{\text{dense}}, p = 0.5)$$

This output feeds into another dense layer:

$$X_{\text{dense}} = \text{ReLU} (X_{\text{flat}} \cdot W_d + b_d)$$

Finally, the output layer consists of 4 neurons with SoftMax activation, providing the probability distribution across the four classes:

$$Y = \text{Softmax} (X_{\text{drop}} \cdot W_o + b_o), \quad \sum_{i=1}^4 Y_i = 1$$

Training mangoNet

In this study, 70% of the images were used for training mangoNet, 20% for validation, and 10% for testing its performance on unseen test images. To prevent data leakage, the images were transferred to the train, validation, and test directories. As¹⁸ suggested, the mangoNet model verification stage should include evaluating the final model on a dataset not seen during training. The study also suggested that ‘test set’ and ‘validation set’ should not be considered the same.

As suggested by⁴⁰, this study applied SMOTE (Synthetic Minority Over-sampling Technique) and Data Augmentation after splitting the dataset into training, validation and testing sets. This ensures that no information from the test set was used during the training phase, which is a critical step to prevent data leakage. SMOTE was applied only to the training set in each fold to balance class distribution. This was done by generating synthetic samples from the existing training data, without affecting the test set. Data augmentation techniques (e.g., random rotations, flips, and brightness adjustments) were applied only to the training data. These transformations are designed to increase the diversity of the training set without introducing any information from the test set.

The mangoNet model was trained using K-fold stratified cross-validation. To ensure that each fold trains the same proportion of class labels as in the original dataset, K-5 stratified cross-validation was selected. In this study, 70% of the images were used for training mangoNet, 20% for validation, and 10% for testing its performance on unseen test images. Since the study utilised two (2) mango leaf disease datasets, Dataset A- the primary dataset and Dataset B- a public dataset, two separate trainings were conducted for each dataset. Hence, test results also emerged from two distinct datasets.

To prevent data leakage, the images were transferred to the train, validation, and test directories. Separating the images into three directories ensured against any potential data leakage, and hence, the test and validation sets were not considered the same. As¹⁸ suggested, the mangoNet model verification stage included evaluating the final model on a test dataset (10% of the images) that was not used during training. As suggested by⁴⁰, this study applied SMOTE (Synthetic Minority Over-sampling Technique) and Data Augmentation after splitting the dataset into training and testing sets. This ensures that no information from the test set was used during the training phase, which is a critical step to prevent data leakage. SMOTE was applied only to the training set in each fold to balance class distribution. This was done by generating synthetic samples from the existing training data, without affecting the test set. Data augmentation techniques (e.g., random rotations, flips, and brightness adjustments) were applied only to the training data. These transformations are designed to increase the diversity of the training set without introducing any information from the test set. The training hyperparameters are provided in Table 2.

Experiments and results

The detailed experimental processes and results are described in the following section:

Results of the mangoNet

The mangoNet model’s fold-wise classification accuracy is shown in Table 3.

The classification report (Table 3) and confusion matrix (Fig. 9) of mangoNet demonstrate the model’s high accuracy. In Fold 1, the model mistakenly classified 2 Healthy as Gall Midge, and 3 Gall Midge instances were misidentified. Folds 3,4, and 5 achieved almost perfect classification and nearly zero misclassification Table 4.

The combined classification metrics for the mangoNet model, as shown in Table 3, demonstrate exceptional performance across all classes. With an overall accuracy of 99.61%, the model achieves balanced precision,

Parameters
Epochs = 50
Batch size = 16
Image size = (64, 64, 3)
Learning rate = 1.0000e-04
K_folds = 5
Optimizer = Adam(learning_rate=Learning rate)
Loss = SparseCategoricalCrossentropy(from_logits=True)
Early stopping = EarlyStopping(monitor='val_accuracy', patience = 10, verbose = 1, restore_best_weights=True) Learning_rate_scheduler = LearningRateScheduler(lambda epoch: LEARNING_RATE * 0.1 ** (epoch//10))
Callbacks = [early_stopping, lr_scheduler]

Table 2. Hyperparameters of training mangoNet.

Fold	Class Name	Precision	Recall	F1-Score	Support
1	Accuracy	99%	99%	99%	1165
	Macro Avg	99%	99%	99%	1165
	Weighted Avg	99%	99%	99%	1165
2	Accuracy	100%			1165
	Macro Avg	100%	100%	100%	1165
	Weighted Avg	100%	100%	100%	1165
3	Accuracy	100%			1165
	Macro Avg	100%	100%	100%	1165
	Weighted Avg	100%	100%	100%	1165
4	Accuracy	100%			1165
	Macro Avg	100%	100%	100%	1165
	Weighted Avg	100%	100%	100%	1165
5	Accuracy	100%			1164
	Macro Avg	100%	100%	100%	1164
	Weighted Avg	100%	100%	100%	1164

Table 3. Classification report of the Fold-wise classification report of the mangoNet model.

recall, and F1 Scores, each at 99.61%. Class-specific results indicate robust performance, with Gall Midge, Powdery Mildew, and Sooty Mould attaining precision, recall, and F1 Scores around 0.99. At the same time, the classifications of all other classes excel with scores nearing 1.00. The macro and weighted averages confirm consistent performance across the dataset, underscoring the model's reliability in mango leaf disease classification. Figure 10 presents the combined confusion matrix.

The fold-wise accuracy and loss curves in Fig. 11 demonstrate the model's ability to converge and remain stable during training. The accuracy curves consistently increase over epochs, with each fold achieving high accuracy by the end of training. This indicates that the model effectively learns the distinguishing features of the diseased classes. The loss curves exhibit a clear downward trend across all folds, indicating a consistent reduction in training and validation loss. This behaviour confirms that the model successfully minimises classification errors during training, achieving stable and efficient learning.

Test data accuracy

Table 5 presents the classification report for the unseen data (10%) of both Dataset A and Dataset B.

Dataset A demonstrates superior consistency and performance across all key metrics (accuracy, precision, recall, F1-score), making it the stronger model overall. Its high accuracy (99%) and balanced precision and recall ensure robust classification. Dataset B, while performing well with strong precision and recall, shows slight performance drops, particularly in recall and F1-score, leading to a slightly lower overall accuracy (97%) compared to Dataset A. However, it remains a strong contender in its own right.

Figure 12 shows the confusion matrixes for two datasets. For Dataset A, the model performs relatively well, with most categories achieving high accuracy—particularly Bacterial Canker, Cutting Weevil, and Die Back, where true positive rates reach 95, 99, and 100, respectively. The model's performance on Dataset A suggests robust classification, as evidenced by the low off-diagonal values. However, in Dataset B, the accuracy is generally lower across most categories, especially for Anthracnose and Bacterial Canker, where the true positives decrease to 80 and 77, respectively.

Benchmark comparison of mangoNet

The mangoNet model was evaluated with several CNN architectures see Table 6 and Figure 13, for results

Table 7 presents the performance of 6 transfer learning methods. MobileNet excels with a high training accuracy of 99%, maintaining strong performance at 85% on validation and 95% on testing, demonstrating excellent generalization. InceptionV3 and Xception also show solid results, with test accuracies of 94% and 92%, respectively. VGG16, however, shows a significant drop in validation and test accuracy, suggesting potential overfitting. ResNet50 struggles across all metrics, with consistently low performance around 41%. DenseNet121 performs well, achieving 92% test accuracy, but slightly underperforms compared to MobileNet and Xception. Overall, MobileNet is the most reliable, while ResNet50 shows the poorest results.

Table 8 presents the performance comparison of two ensemble models. The ensemble models were tested using three voting models (Soft Voting, Rank-Based, and Hard Voting). The Soft Voting model consistently outperformed the other models in both datasets, with a DIX of 94%, a recall of 94%, an F1 score of 94%, and RIV accuracies of 89%, precision of 89%, recall of 89%, and F1 score of 89%. Rank-based voting was the least effective, particularly on the RIV dataset. Hard Voting demonstrated a balanced performance across metrics but did not surpass Soft Voting in either dataset. These findings suggest that, for the given datasets, Soft Voting is the most reliable model, while Rank-Based voting is less effective and may require further tuning to improve performance.

Disease Name = Short abbreviation

Anthracnose = AN Bacterial Canker =BC Cutting Weevil = CV Die Back =DB

Gall Midge = GM Healthy= H Powdery Mildew =PM Sooty_Mould= SM

Fold 1

	AN	BC	CW	DB	GM	H	PM	SM
AN	147	0	0	0	0	0	0	0
BC	0	152	0	0	0	1	0	0
CV	0	0	157	0	0	0	0	0
DB	0	0	0	155	0	0	0	0
GM	0	1	0	0	114	0	0	2
H	0	0	0	0	2	137	0	0
PM	0	0	0	0	1	0	144	3
SM	0	0	0	0	0	0	1	148

Fold 2

	AN	BC	CW	DB	GM	H	PM	SM
AN	129	0	0	0	0	0	0	0
BC	0	153	0	0	0	0	0	0
CV	0	0	136	0	0	0	0	0
DB	0	0	0	141	0	0	0	0
GM	0	1	0	0	167	0	1	0
H	0	0	0	1	0	139	0	0
PM	0	0	0	0	0	0	150	0
SM	0	0	0	0	0	0	2	145

Fold 3

	AN	BC	CW	DB	GM	H	PM	SM
AN	151	0	0	0	0	0	0	0
BC	0	142	0	0	1	0	0	0
CV	0	0	147	0	0	0	0	0
DB	0	0	0	145	0	0	0	0
GM	0	0	0	0	146	1	0	1
H	0	0	0	0	0	135	0	0
PM	0	1	0	0	0	0	155	0
SM	0	0	0	0	0	0	0	140

Fold 4

	AN	BC	CW	DB	GM	H	PM	SM
AN	158	0	0	1	0	0	0	0
BC	0	130	0	0	0	0	0	0
CV	0	0	148	0	0	0	0	0
DB	0	0	0	144	0	0	0	0
GM	0	0	0	0	142	0	0	0
H	0	0	0	0	0	151	0	0
PM	0	0	0	0	0	0	145	0
SM	0	0	0	0	0	0	0	146

Fold 5

	AN	BC	CW	DB	GM	H	PM	SM
AN	141	0	0	0	0	0	0	0
BC	0	169	0	0	0	0	0	0
CV	0	0	139	0	0	0	0	0
DB	0	0	0	124	0	0	0	0
GM	0	0	0	0	162	0	0	0
H	0	0	0	0	0	155	0	0
PM	0	0	0	0	0	0	129	0
SM	0	0	0	0	0	0	0	143

Fig. 9. Fold-wise confusion matrix of mangoNet on test data.**Computational cost of the mangoNet and deep learning models**

Since this study aims to develop a lightweight CNN for smart mango orchards and the model will be integrated into an edge device, computational cost is also an important metric. The model structure and computational cost are presented in Table 9, 10 and 11. The analysis is essential to understand the suitability of mangoNet for edge devices.

The parameter settings for mangoNet and other CNNs are presented in Table 9. The table compares the total, trainable, and non-trainable parameters for mangoNet and six (6) SOTA CNNs. MobileNetV2 had the fewest

Combined Accuracy: 99.61%
Combined Precision: 99.61%
Combined Recall: 99.61%
Combined F1-Score: 99.60%

Table 4. Combined results of the mangoNet model.

	AN	BC	CW	DB	GM	H	PM	SM
AN	726	0	0	1	0	0	0	0
BC	0	746	0	0	1	1	0	0
CV	0	0	727	0	0	0	0	0
DB	0	0	0	709	0	0	0	0
GM	0	2	0	0	731	1	1	3
H	0	0	0	3	2	717	0	0
PM	0	1	0	0	1	0	723	3
SM	0	0	0	0	0	0	3	722

Fig. 10. Combined confusion matrix of mangoNet.

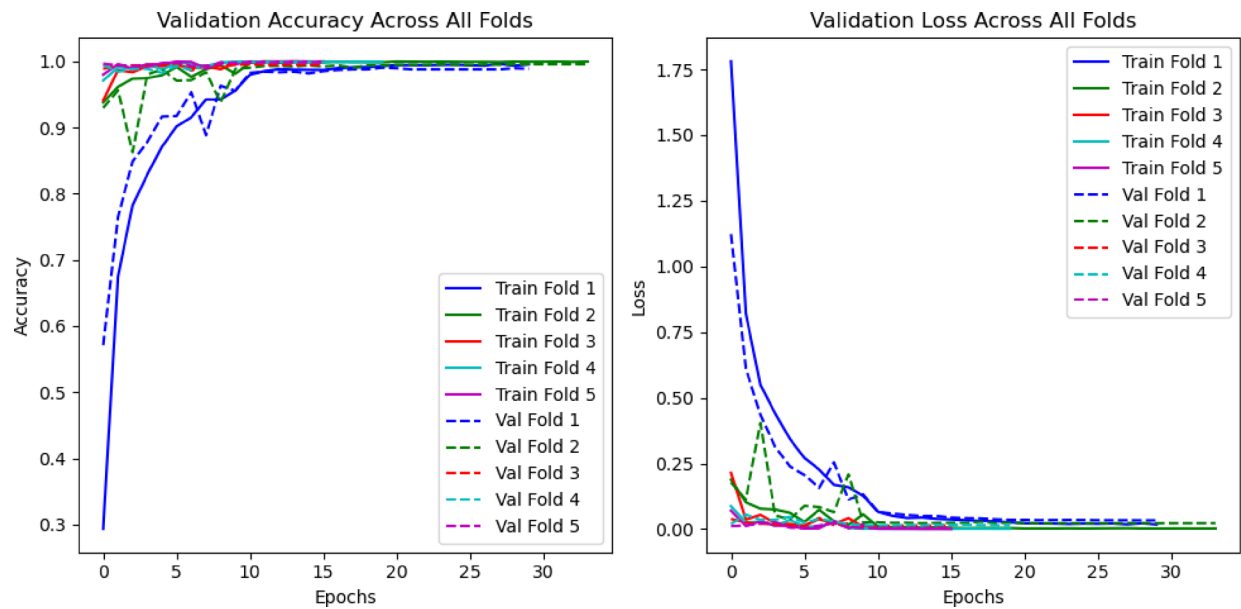


Fig. 11. Fold-Wise Classification Report with Epochs and Accuracy.

total parameters; however, its performance was 78%. However, our developed mangoNet has more parameters than MobileNetV2 but fewer than other CNNs. In most modern CNN architectures, a minimal number of non-trainable parameters is expected.

Table 10 presents the total training times and epoch statistics for mangoNet and SOTA CNNs. The data mangoNet, with the shortest total training time of 285.84 s, took 26 epochs, showing the fastest epoch time

	Precision	Recall	F1-Score	Support	Precision	Recall	F1-Score	Support
Anthraco	99%	100%	0.99	82	100%	92%	96%	50
Bacterial C	99%	99%	99%	76	98%	98%	98%	50
Cutting Wee	100%	100%	100%	85	100%	100%	100%	50
Die Back	100%	100%	100%	73	100%	100%	100%	50
Gall Midge	95%	96%	95%	72	96%	98%	97%	50
Healthy	100%	99%	99%	83	96%	100%	98%	50
Powdery Mild	96%	93%	94%	81	94%	92%	93%	50
Sooty Mould	94%	97%	96%	88	90%	94%	92%	50
Accuracy				640			97%	400
Macro Avg	98%	98%	98%	640	97%	97%	97%	400
Weighted Avg	98%	98%	98%	640	97%	97%	97%	400
Classification Report (Dataset A)					Classification Report (Dataset B)			

Table 5. mangoNet model accuracy on test data (Classification report).

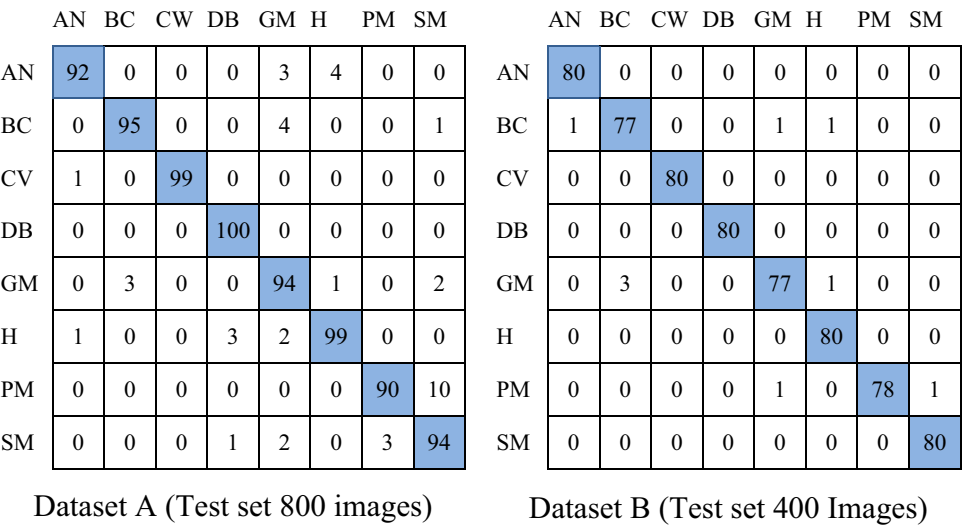


Fig. 12. mangoNet test performance (Confusion matrix).

Network Type	Model	Accuracy
Spatial exploitation-based	VGG16	98%
Multipath	DenseNet121	99%
Depth-based multi-connection	InceptionV3	97%
Multipath	Resnet 50	97%
width-based multi-connection	Xception	97%
depth-wise separable convolutions	MobileNetV2	78%

Table 6. Performance comparison of SOTA CNNs.

mean (10.97s) with very low variability (0.12s). In short, mangoNet was the fastest and least variable, while InceptionV3 and VGG16 were the slowest but still notable for their stable epoch times.

Table 11 presents computational cost matrices of mangoNet and other SOTA CNNs. The data suggests that mangoNet is the smallest and most efficient, both in terms of size and inference time. mangoNet is the smallest at 76 MB, with an inference latency of 71 ms/image, requiring 6 GFLOPs and the least memory for training (2111.37 MB. DenseNet121 has the largest model size at 961 MB, with a high inference latency of 132 ms/image and train memory usage of 2777.14 MB. It also requires 124 GFLOPs, indicating high computational demand. VGG16, at 82 MB, has lower inference latency (74 ms/image) and similar FLOP requirements to ResNet50 (6 GFLOPs).

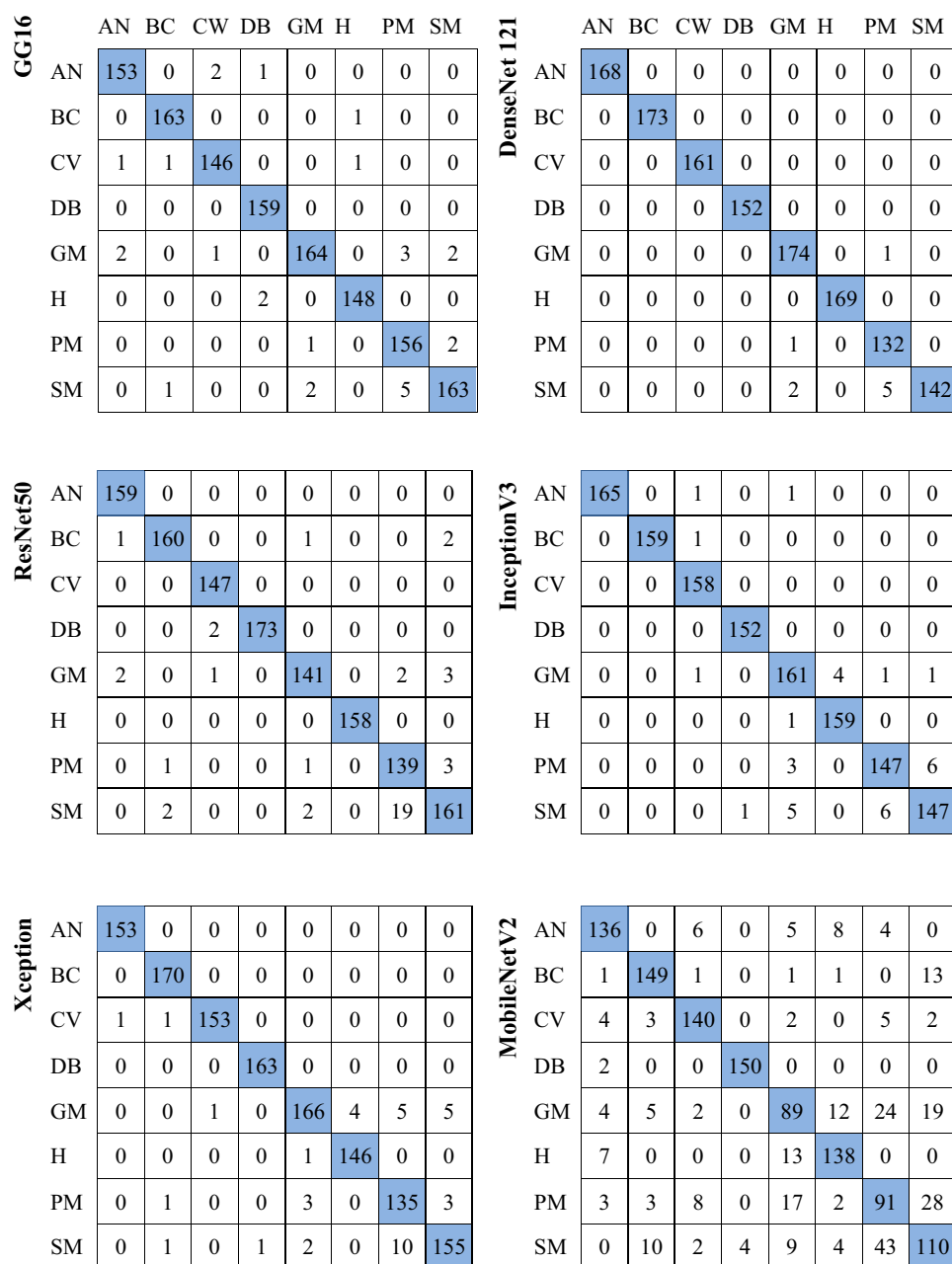


Fig. 13. Confusion matrix of SOTA CNNs.

Model	Training Accuracy	Validation Accuracy	Test Accuracy
VGG16	81%	72%	68%
MobileNet	99%	85%	95%
Xception	97%	82%	92%
InceptionV3	98%	83%	94%
ResNet50	41%	41%	41%
DenseNet121	95%	82%	92%

Table 7. Performance comparison of the Transfer Learning.

DIX	Soft Voting	Accuracy	94%
		Precision	94%
		Recall	94%
		F1 Score	94%
	Rank-Based	Accuracy	81%
		Precision	86%
		Recall	82%
		F1 Score	79%
	Hard Voting	Accuracy	82%
		Precision	89%
		Recall	82%
		F1 Score	82%
RIV	Soft Voting	Accuracy	89%
		Precision	89%
		Recall	89%
		F1 Score	89%
	Rank-Based	Accuracy	57%
		Precision	78%
		Recall	57%
		F1 Score	48%
	Hard Voting	Accuracy	88%
		Precision	88%
		Recall	88%
		F1 Score	86%

Table 8. Performance comparison of the Ensemble model.

Model	Total Parameters	Trainable Parameters	Non-Trainable Parameters
ResNet50	24,113,284	24,060,164	53,120
InceptionV3	22,328,356	22,293,924	34,432
Xception	21,387,052	21,332,524	54,528
VGG16	14,847,044	14,847,044	0
DenseNet121	7,300,932	7,217,284	83,648
MobileNet	3,492,292	3,470,404	21,888
mangoNet	3,987,400	3,987,400	0

Table 9. mangoNet and SOTA CNNs' parameter comparison.

Model	Total Train Time (s)	Epoch Time Mean (s)	Epoch Time Std (s)	Epochs Used
InceptionV3	1657	57	0.47	29
VGG16	1519	84	0.47	18
ResNet50	904	27	0.74	33
Xception	661	27	0.74	24
MobileNet	710	19	0.53	37
DenseNet121	385	9	0.39	20
mangoNet	285	10	0.12	26

Table 10. Model training time of each (seconds).

FNR macro-average of the mangoNet model

The FNR curve (see Fig. 14) suggests that mangoNet demonstrates relatively low FNR values across most classes, especially for AN (0.0014) and CV (0.0). However, the FNRs for GM (0.0095) and PM (0.0069) are higher than those of other models. VGG16 exhibits higher FNR values than MangoNet, particularly for BC (0.0468) and GM (0.0465).

Model	Model Size (MB)	Inference Latency (ms/image)	Inference Memory (MB)	FLOPs (G)	GFLOPs (Approx)	Train Memory (MB)
DenseNet121	961	132	6	124	124	2777
Xception	109	93	2	80	80	3082
MobileNet	109	83	2	80	80	3078
InceptionV3	109	47	3	80	80	2944
ResNet50	82	99	4	7	7	2804
VGG16	82	74	2	6	6	3128
mangoNet	76	71	2	6	6	2111

Table 11. mangoNet and SOTA CNNs' model comparison of computational efficiency.

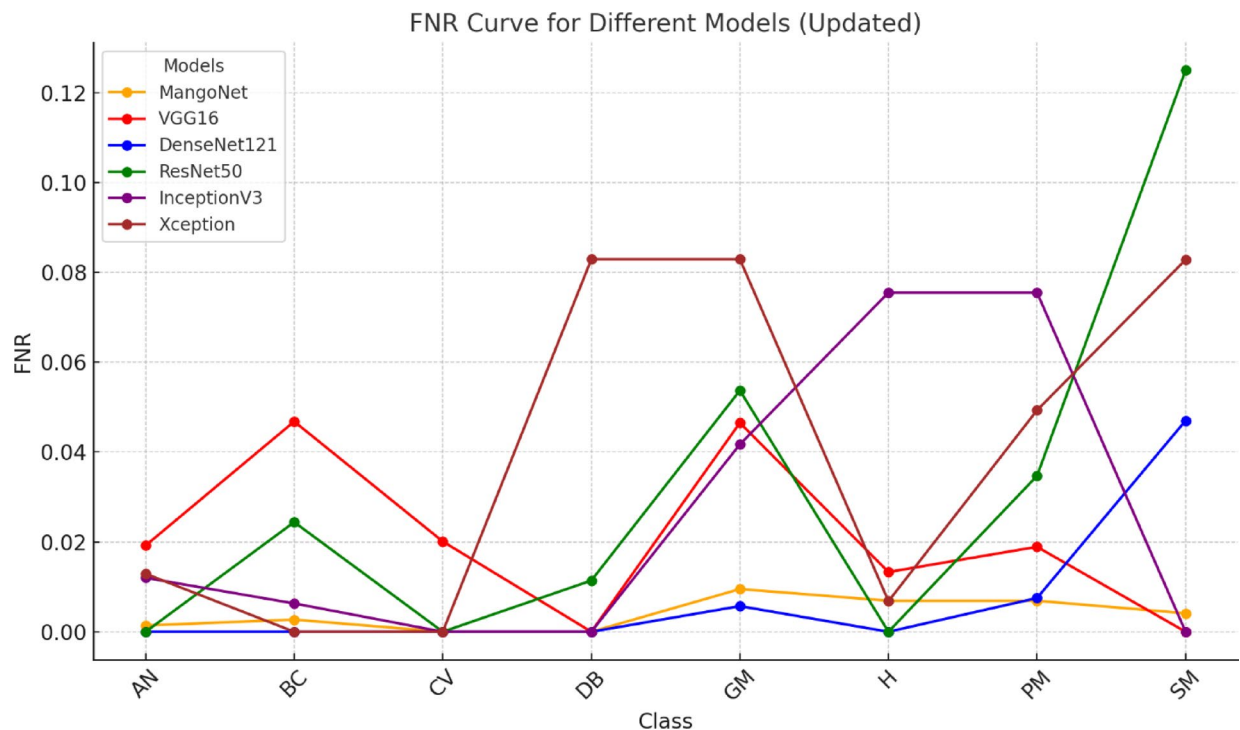


Fig. 14. FNR macro-average.

Descriptive statistics analysis

The descriptive statistics suggest that mangoNet outperforms other SOTA CNNs (see Tables 12 and 13). mangoNet achieves a high mean accuracy of 91%, which is competitive and very close to top performers such as ResNet50 and Inception V3. In addition to its strong accuracy, MangoNet exhibits a relatively low standard deviation (15) in accuracy, indicating consistent performance across different runs. The confidence interval for MangoNet's accuracy (85% to 97%) is also relatively tight, showing that its performance is both high and stable.

Result of the XAI

This study utilised three explainable AI techniques, LIME, SHAP, and Grad-CAM, to generate local and global explanations for the mangoNet. The inclusion of explainable AI (XAI) techniques such as SHAP, LIME, and Grad-CAM adds transparency to the model's decision-making process. These techniques enable us to identify which features of mango leaves were most influential in predicting the class label. SHAP values were used to explain the model's predictions, highlighting which parts of the mango leaf (e.g., lesions, edges) contributed most to the classification. For example, Anthracnose lesions were predominantly classified based on color variations and texture. LIME was applied to interpret individual predictions, helping to understand which regions of the image (e.g., the edges of the Powdery Mildew lesion) were decisive for the model. Grad-CAM visualizations were used to generate heatmaps over the images, showing where the model focused its attention during predictions. This approach is beneficial for demonstrating how the model handles diseased regions in a visual and interpretable manner. These findings enhance the paper's impact by addressing concerns about model transparency and interpretability, particularly for end users such as farmers and agricultural experts.

Metric	MangoNet	VGG16	DenseNet121	Inception V3	ResNet50	Xception	MobileNet
Acc Mean	91.3%	80.5%	82.1%	92.6%	93.3%	91.3%	87.5%
Acc Std	15.4%	20.4%	18.7%	9.4%	9.0%	9.7%	16.7%
Acc CI Low	84.9%	70.1%	73.1%	89.0%	90.1%	87.1%	81.9%
Acc CI High	97.6%	91.0%	91.1%	96.3%	96.6%	95.5%	93.2%
Loss Mean	23.0%	48.5%	45.2%	20.2%	18.2%	23.7%	33.7%
Loss Std	38.5%	47.8%	43.5%	24.4%	23.4%	25.3%	43.7%
Loss CI Low	7.1%	24.1%	24.3%	10.7%	9.8%	12.8%	18.9%
Loss CI High	38.9%	72.9%	66.1%	29.6%	26.6%	34.6%	48.5%
Acc Cohen's d	1.93	1.95	1.38	3.84	3.33	2.39	1.25
Acc Hedges' g	1.74	1.77	1.20	3.57	3.08	2.07	1.14
Val Acc	98.9%	93.8%	95.6%	99.1%	99.5%	98.6%	93.1%
Test Acc	98.8%	93.6%	96.1%	99.4%	99.2%	98.6%	90.6%

Table 12. Statistical analysis of models.

Model	Mean	Std	Median	IQR	Range
DenseNet121	36.9%	29.7%	24.8%	27.5%	98.8%
ResNet50	185.5%	244.9%	76.5%	145.0%	870.5%
InceptionV3	72.9%	47.0%	55.8%	74.9%	167.2%
Xception	49.5%	34.8%	38.0%	36.0%	110.7%
MobileNet	73.6%	52.3%	59.2%	72.0%	170.3%
VGG16	207.9%	0.1%	207.8%	0.0%	0.2%
MangoNet	52.6%	49.3%	30.5%	53.3%	169.1%

Table 13. Loss calculation of mangoNet and SOTA CNNs.

Visualisation using LIME and SHAP

The LIME prediction was generated using a mango.h5 model (mangoNet architecture + Weights) that was saved after the training process. The input images were resized (56 × 56 pixels) and normalised pixel values to a [0, 1] range before passing them to the model for prediction. The Red and Green Marks generated using LIME explain the input image, thus assisting us in understanding the determining process of mango leaf diseases (see Fig. 15).

SHAP explanations are generated using a pre-trained mango.h5. The SHAP explainer (shap.Explainer) is initialized with the model and masker to compute Shapley values (see Fig. 16). The red regions indicate features that positively influence the prediction of a class, while the blue areas indicate features that reduce the likelihood. The colour-coded SHAP values quantify the importance of these regions, offering transparency and interpretability essential for medical AI applications.

Grad-CAM analysis of classified mango leaf disease detection

Figure 17 (A, B) highlights the Grad-CAM visualization and the model's incorrect predictions for various mango leaf disease cases. The red color on the maps denotes greater attention paid to those locations, while the blue color denotes less attention to those regions. The heatmaps reveal that the model often focused on irrelevant or non-diseased areas, leading to misclassifications.

Figure 18 (A) shows the misclassified images; the Grad-CAM clearly highlights the non-diseased region, demonstrating the model's incorrect focus on relevant features for its prediction. Figure 18 (B) shows the correctly classified images where Grad-CAM clearly highlights the diseased area.

Pixel intensity of classified mango leaf disease detection

Pixel intensity visualization (See Fig. 18A and B) uses a Gradient × Input Explanation to illustrate the model's correct/incorrect classifications. In this case, the highlighted regions in the Gradient × Input explanation do not align well with the actual disease, indicating the model focused on irrelevant areas and missed critical disease features. This suggests a misalignment between the model's learned features and the original leaf disease.

Pixel Intensity Analysis: Correct/misclassified images

This boxplot (Fig. 19) compares the pixel-intensity distributions of misclassified and correctly classified instances, with descriptive statistics. Correctly classified images (orange) show a lower mean pixel intensity of 154.88 compared to 174.94 for misclassified images (blue), with a broader range of values (64.82–229.66 vs. 93.86–224.61) (for more details, see Table 14). The median intensity for correctly classified cases is 158.54, which is significantly lower than the 182.38 for misclassified instances. The interquartile range (IQR) is also larger (133.15–181.02 vs. 156.29–198.99), suggesting that images with distinct and lower pixel intensities are more likely to be correctly classified. Misclassified images exhibit lower variability (std = 35.56 vs. 30.16) and narrower

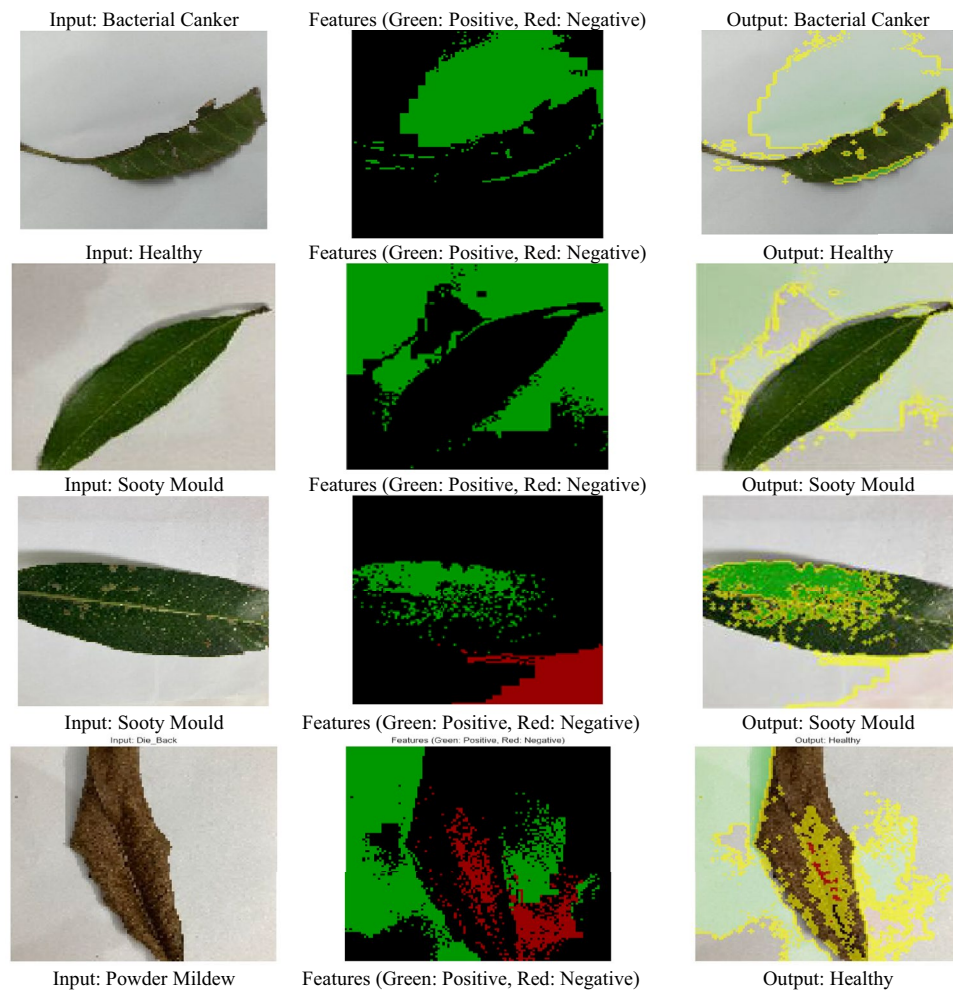


Fig. 15. LIME partition explainer of mango leaf images.

intensity ranges, indicating that the model struggles when features are less prominent or less variable, which leads to errors. This highlights the importance of distinct intensity contrasts for accurate model predictions.

Pixel intensity is an important feature for image classification⁴¹. From the boxplot of Dataset A, the “Correctly classified” class shows a relatively tight range of pixel intensities. The median value is closer to the upper end of the range. This suggests that the dataset has more consistent, distinct pixel intensities for correctly classified instances. This makes it easier for mangoNet to classify the classes. However, in Dataset B (97% Accuracy), the boxplot shows a broader range of pixel intensity values, particularly for misclassified instances. The wider distribution, especially at lower values, indicates greater variation in pixel intensities between correctly and misclassified instances. This overlap may make it difficult for the model to distinguish the classes. Moreover, Dataset A contained 800 images per class, totaling 6400. On the other hand, dataset B had 500 images per class, totaling 4000 Images.

Pixel intensity of TP-FP-TN-FN for mango leaf disease

Furthermore, the pixel intensities of 10 images, including True Positive, True Negative, False Positive, and False Negative, were analysed (Fig. 20; Table 15). The True Negative class has the highest mean pixel intensity (159.93) and a broader range (IQR = 49.5). This suggests that image detection error arises from highly variable or extreme intensities (max = 204.25). The True Positive class (mean = 156.19, IQR = 43.5) exhibits more consistent distributions, with medians of 162.00 and 164.50, respectively. This suggests that the model performs well when intensities align with specific patterns. However, False Positives (mean = 156.40, IQR = 59.0) and False Negatives (mean = 157.23, IQR = 71.0) exhibit greater variation, indicating that the model struggles most when pixel intensities are extreme or highly variable.

Smart IoT and mangoNet-based precision agriculture

mangoNet was developed to deploy a smart monitoring system for mango orchards. In this proposed system, the following components are required: an IoT camera, an IoT Gateway, a machine learning server (MLS), and an IoT router (see Fig. 21). The MLS serves as the central hub of the system. The trained ML model resides in

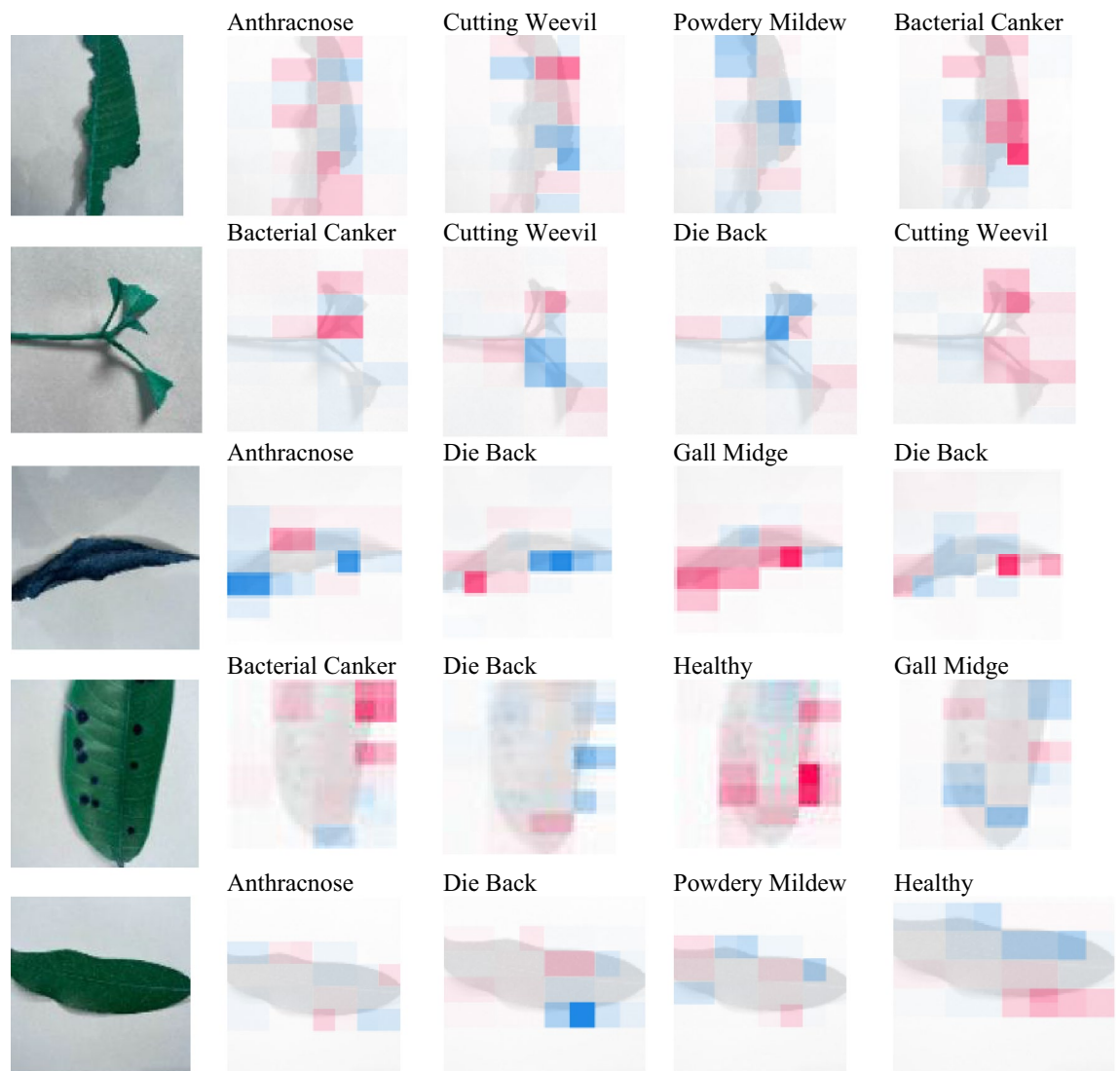


Fig. 16. SHAP explainer of mango leaf images.

the MLS. The system proposes using a wireless connection to connect all devices. The components are discussed below:

Mango leaf disease images are input into the system via IoT Cameras installed in orchards or handheld imaging devices. The system supports wireless data transmission from portable imaging devices, enabling real-time processing in resource-constrained settings. Farmers or users can upload photos directly through a connected web interface or via external devices, such as USB drives.

The trained mangoNet model will reside in MLS. It will collect data from image processing and evaluate whether any diseased leaves are present. If it detects any disease on the mango leaf, it will propagate to the user interface. A secure IoT connector connects the MLS to users' edge devices, such as computers, laptops, or mobile phones. The proposed mangoNet model was integrated into both SteamLit and Android applications to test its performance. Example results are shown in Figs. 22 and 23.

The mangoNet application was tested on a Realme Note 70 mobile phone as an edge device. The smartphone features a 6300mAh battery, a 90 Hz HD+ display, an IP54 dust- and water-resistant design, a Unisoc T7250 chip, and a 13MP primary camera. The mangoNet achieved a modest 1 FPS, indicating that the system processes one frame per second. The device's battery consumed 21% during operation. The temperature data showed that the CPU temperature did not increase significantly, as the data was recorded as "N/A." This suggests that the application did not impose a substantial thermal load on the device, even with continuous operation.

Discussion

This research aimed to present a lightweight CNN suitable for IoT devices. Unlike other studies^{45–72}, which provided only machine learning models, this study adopted a practical approach by integrating mangoNet into the edge device. The main characteristics, therefore, of the lightweight CNN are limited parameters, non-intricate, and straightforward network architectures. Moreover, the model must provide high accuracy. The authors noted

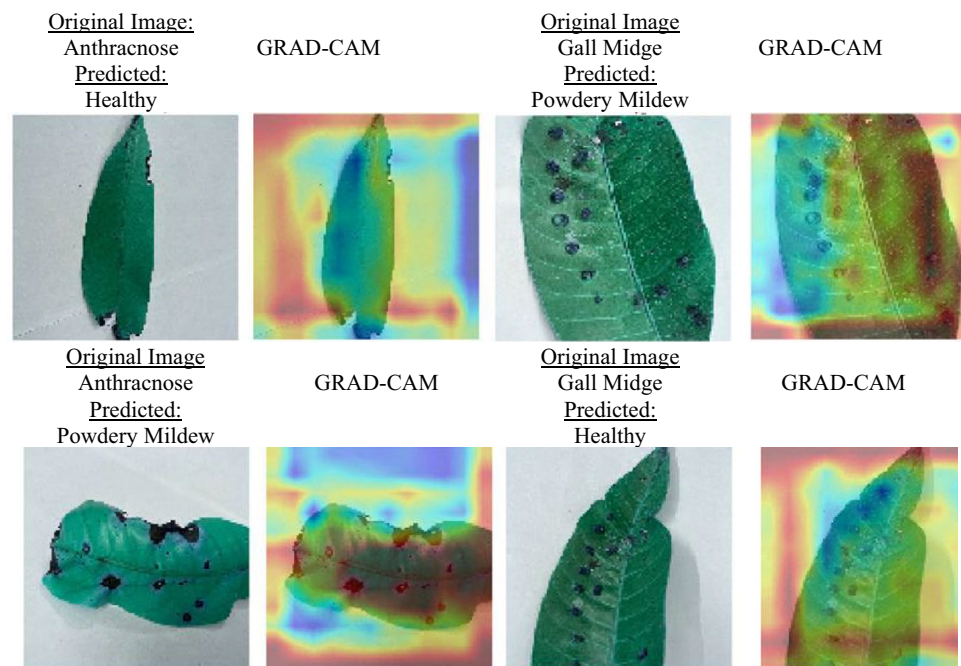


Fig. 17. (A): GRAD-CAM view of misclassified images.

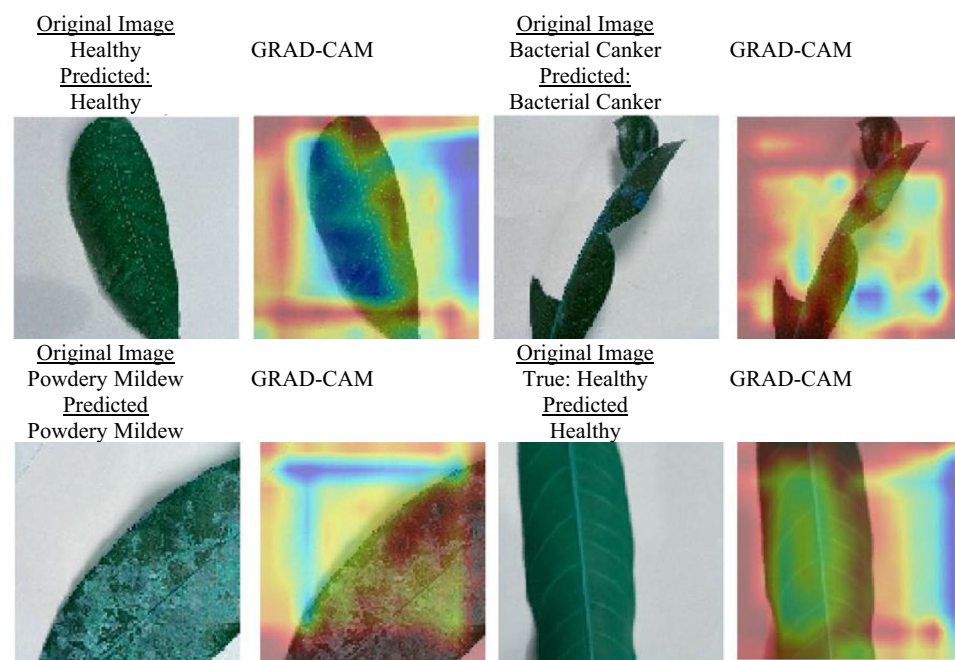


Fig. 17. (B): Grad-CAM view of correctly classified images.

that the accuracy of machine learning models is insufficient, as they have shown poor performance in real-world settings⁴². With only 3,987,400 parameters and 99% accuracy on unseen test images, mangoNet is a suitable model for a smart mango orchard. The findings suggest that the proposed mangoNet effectively classifies mango diseases using leaf images and achieves the highest accuracy (99.61%) compared to SOTA CNN models.

The mangoNet was trained and evaluated using three Python data generation techniques: TensorFlow/Keras, flow_from_directory, flow_from_dataframe, and flow (used with NumPy arrays). It was evident that flow_from_directory provides better results than the other two data pipelines. The flow_from_directory function efficiently loads images batch by batch, reducing memory usage compared to loading the entire dataset into RAM using flow_from_dataframe and flow (which is typically used with NumPy arrays). Adam remains a stable performer,

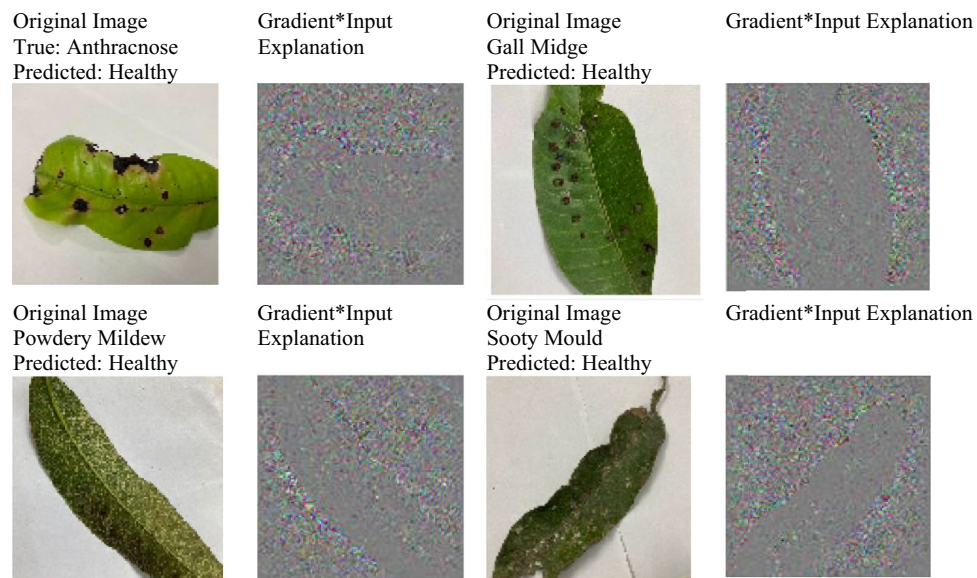


Fig. 18. (A): Pixel intensity of misclassified images.

while RMSprop fluctuates across models, underscoring the importance of carefully selecting an optimizer based on architecture-specific convergence behavior.

Among the SOTA CNNs, DenseNet121 and VGG16 achieved the best accuracies of 99% and 98%, respectively (see Table 4; Fig. 24). InceptionV3, ResNet50, and Xception achieved 97% accuracy across most classes. However, MobileNetV2 performed the least, with 78% accuracy. Table 4 shows the classification report of SOTA CNNs. In this study, we have used MobileNet's baseline architecture. Although MobileNet is often described as a lightweight CNN, its depth and width reduction techniques, along with image compression, result in lower classification accuracy³⁸. The study³⁹ discusses the limitations of MobileNetV2, particularly its suboptimal feature extraction compared to other lightweight architectures. Therefore, studies such as^{26,43,44} improved MobileNet for leaf disease detection. Among SOTA CNNs, the DenseNet architecture is more efficient in detecting and classifying mango leaf disease. Our findings are also supported by^{28,36}. The study by²⁸ Applied DenseNet121 and achieved the best result (99%) for mango leaf disease identification. The study by³⁶ argued that the DenseNet architecture uses densely connected skip connections. The DenseNet connection pattern connects all layers directly, ensuring maximum information flow throughout the network. The feed-forward nature for each layer is maintained by receiving input from the previous layers. However, the authors have already noted the challenges posed by gradient-based loss functions in leaf image analysis^{37,45}. In agricultural datasets, the imbalance class is also a common challenge for loss functions. In summary, DenseNet is a well-performing CNN model in which all layers are connected to the network.

Lastly, this study applied three (3) XAI techniques, LIME, SHAP, and Grad-CAM to expand our understanding of mangoNet's decision-making process. The pixel-intensity analysis using a statistical method contributes to the growing literature on using XAI to enhance image analysis and diagnosis. It provides insights into the interpretability and transparency of CNN for mango leaf disease classification.

Contribution

The mangoNet was developed for an IoT-based mango orchard. The model was developed to detect mango leaf disease, and its accuracy in classifying two mango leaf datasets positioned it as a suitable lightweight model. The accuracy comparison with SOTA CNNs, transfer learning and ensemble models supports its acceptability over other variants. The acceptability of mangoNet is also achieved by including extensive computational cost analysis and statistical analysis. Lastly, the integration of XAI and in-depth analysis of pixel intensity of correctly classified and misclassified mango leaves has expanded the use of XAI techniques. The study articulates its objectives, encompassing model development, constraints, and performance evaluation relative to benchmark works. The novelty of mangoNet lies in its balanced sequential architecture with 4×4 kernels and progressive filter expansion, enabling early layers to capture low-level features and deeper layers to extract high-level features. With only 3,987,400 Total parameters, the mangoNet integrated website and mobile application is expected to empower farmers, agricultural departments, and farming stakeholders, enabling deployment in farming environments to support the diagnosis of mango leaves.

Conclusion and future work

This study presents a lightweight CNN, mangoNet, for smart mango orchards, enabling real-time disease monitoring. The results show that the lightweight mangoNet model outperforms 6 SOTA CNNs on a dataset of 7023 mango leaf images and is therefore more applicable in an IoT environment. By utilizing XAI techniques such as LIME, SHAP, and Grad-CAM, we gain insight into the model's decision-making process. Additionally,

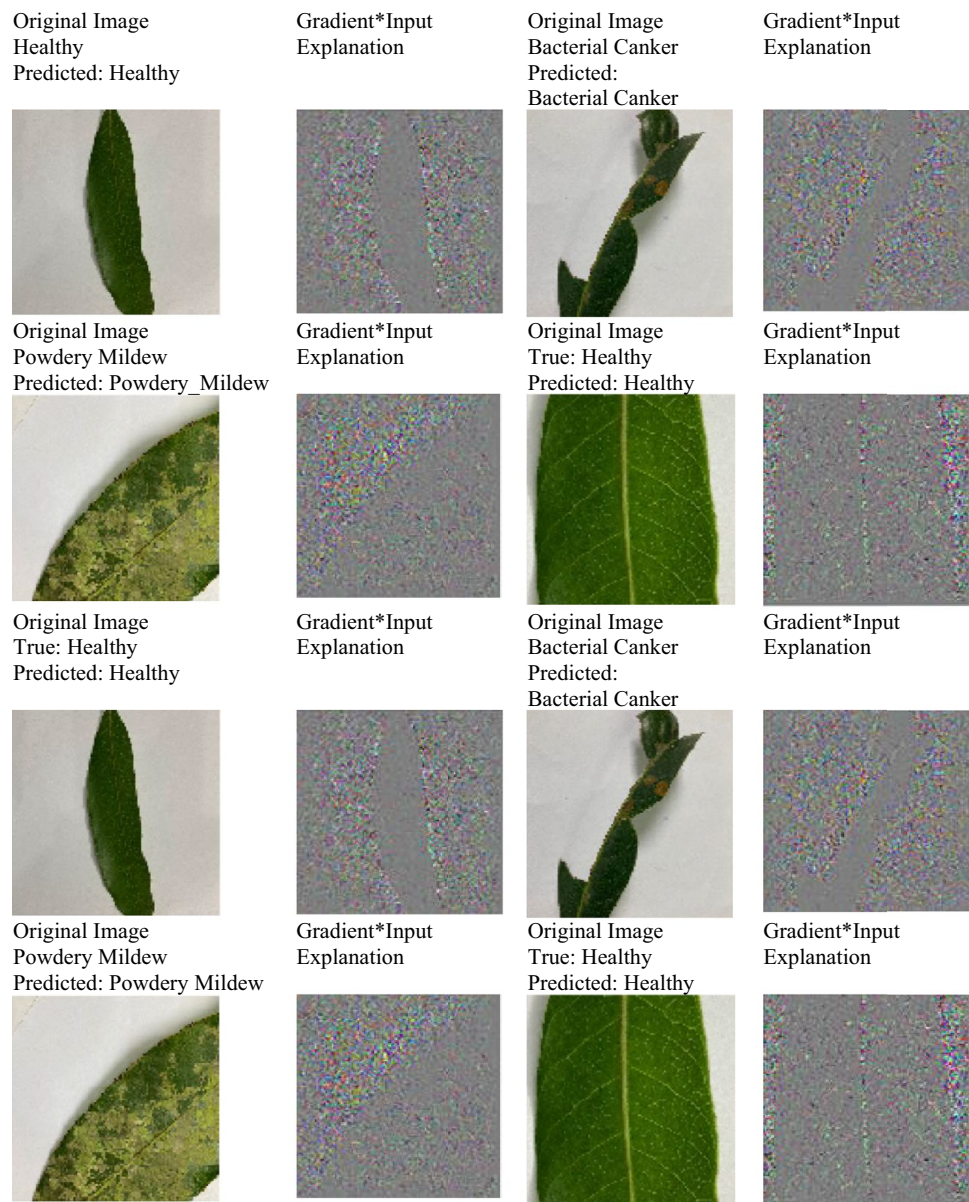


Fig. 18. (B): Pixel intensity for correctly classified images.

pixel-intensity analysis played a vital role in understanding how the model distinguished between healthy and diseased leaves, thereby contributing to accurate classification.

Although this study makes several contributions, it also has some limitations. The primary limitation is that the model was trained on a custom dataset, which may still contain inherent biases in data collection and annotation. Secondly, the model was tested only on mango leaf images, limiting its generalizability to other plant species and disease types. In the future, the model should be applied to a broader range of plant diseases across different crops to assess its robustness. Additionally, the experimental results were not validated by plant pathology experts; instead, the results were based solely on computational evaluations. The results were tested on the unseen dataset, and the test results were manually confirmed to match the class. Furthermore, the outcome of the implemented Android APP was validated with the orchard owner, who had significant experience in mango farming. However, the outcome was not validated in a real-world setting, as no organization funded the study to purchase the necessary IoT infrastructure. Lastly, an agriculturist from the university confirmed the validity of the test result and the detection using the mangoNet APP. Future studies should incorporate expert validation to enhance the reliability of the findings. Furthermore, this study did not explore transfer learning; instead, it focused on training a lightweight CNN from scratch. Future research should investigate whether pre-trained models can further improve mango leaf disease classification. Finally, this study used Python (TensorFlow/Keras) for implementation; future work could explore different programming languages and frameworks to optimize performance and scalability.

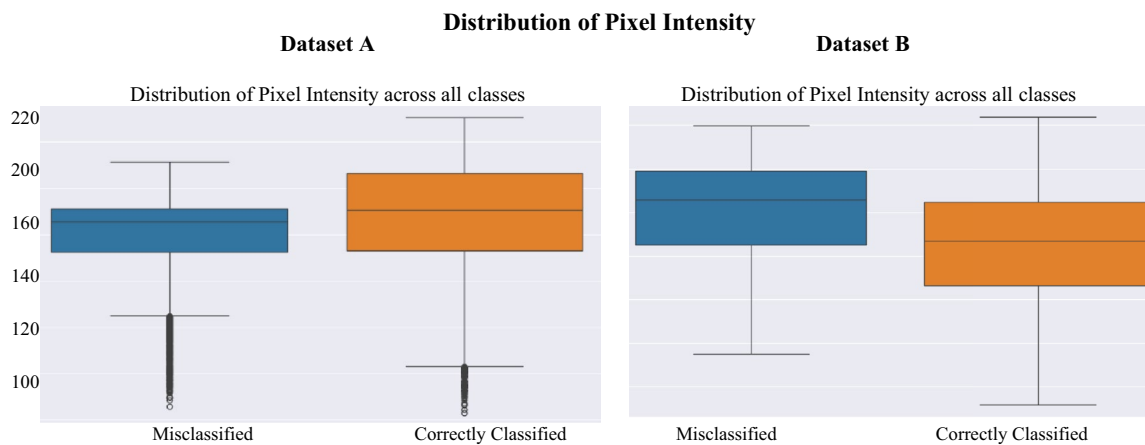


Fig. 19. Distribution of Pixel intensity of misclassified/correctly classified images.

Statistic	Dataset A		Dataset B	
	Misclassified	Correctly Classified	Misclassified	Correctly Classified
Count	12,544	12,544	4096	4096
Mean	175	155	185	181
Std	30	36	30	24
Min	94	65	100	107
25%	156	133	168	168
50%	182	159	192	185
75%	199	181	206	199
Max	225	230	228	223
Interquartile Range	43	48	37	31

Table 14. Descriptive Statistics for Pixel Intensity of correctly classified and misclassified images.

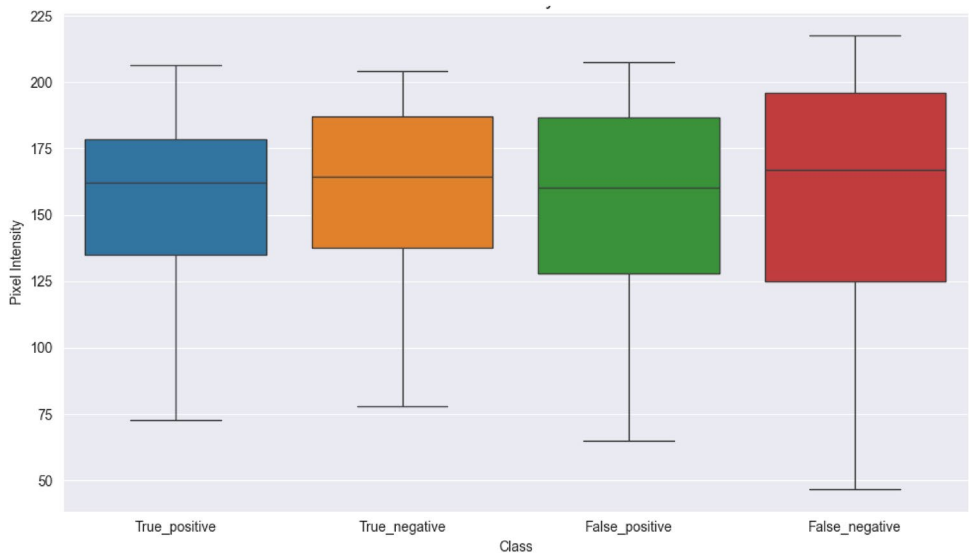


Fig. 20. Distribution of Pixel intensity of TP-TN-FP-FN.

In conclusion, our investigation demonstrates the effectiveness of CNN in detecting and classifying mango leaf diseases. Our model achieved high accuracy, precision, recall, and F1 scores on the test set, validating its performance. Additionally, LIME and SHAP visualizations provided both local and global explanations for the model’s predictions, highlighting critical regions for disease classification. Moving forward, we identify two

Statistics	True Positive	True Negative	False Positive	False Negative
Count	12,544	12,544	12,544	12,544
Mean	156.1875	159.9319	156.3957	157.2292
Std	30.69588	29.85484	32.87967	44.86355
Min	72.625	78.125	64.875	46.875
25%	134.875	137.625	127.875	124.875
50%	162	164.5	160.375	166.875
75%	178.375	187.125	186.875	195.875
Max	206.25	204.25	207.375	217.5
Interquartile Range	43.5	49.5	59	71

Table 15. Descriptive Statistics for Pixel Intensity of TP-TN-FP-FN images.



Fig. 21. Real-time mango leaf disease monitoring architecture.

key directions for future research: expanding the dataset to include a wider range of environmental conditions and exploring advanced deep learning architectures to improve generalization. Overall, our study presents a promising approach to a smart mango orchard for real-time leaf disease detection, and we hope our findings encourage further research in this important domain.

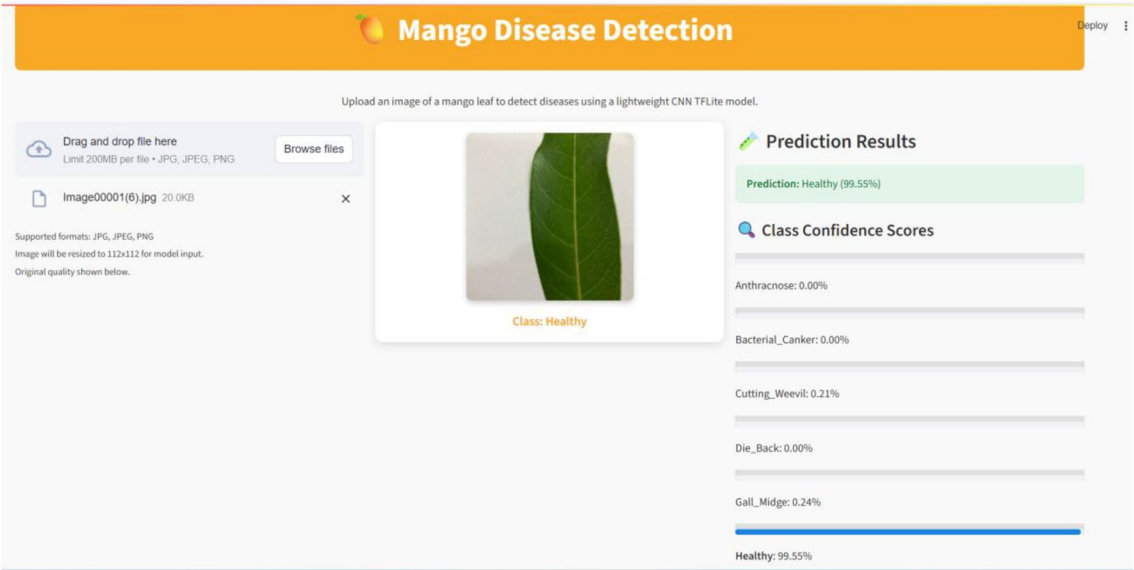


Fig. 22. (A): High-level architecture of the website and mobile app.

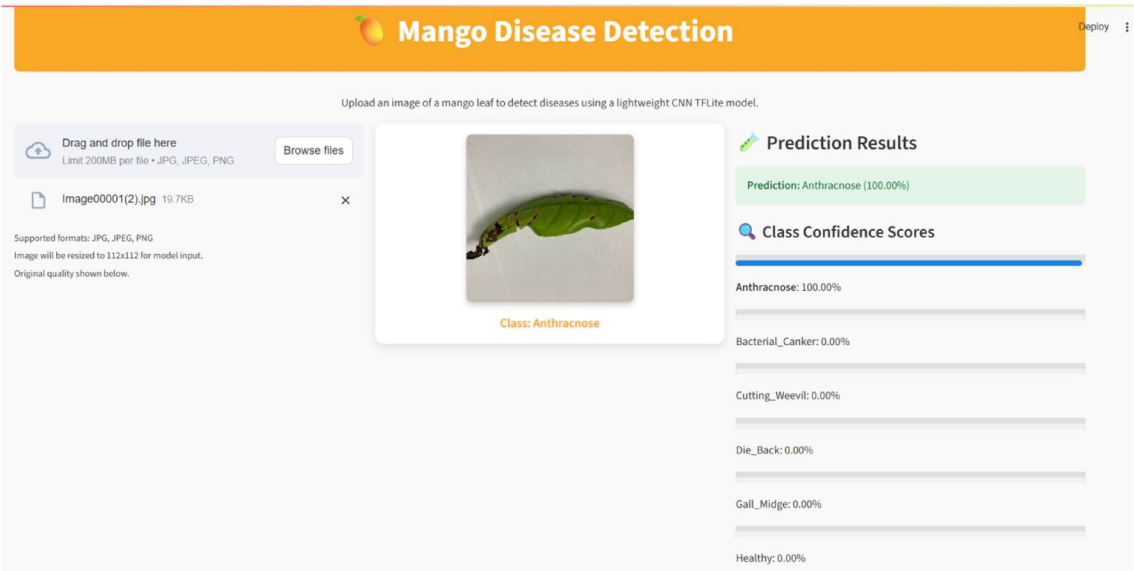


Fig. 22. (B): StreamLit web application interface and sample result.

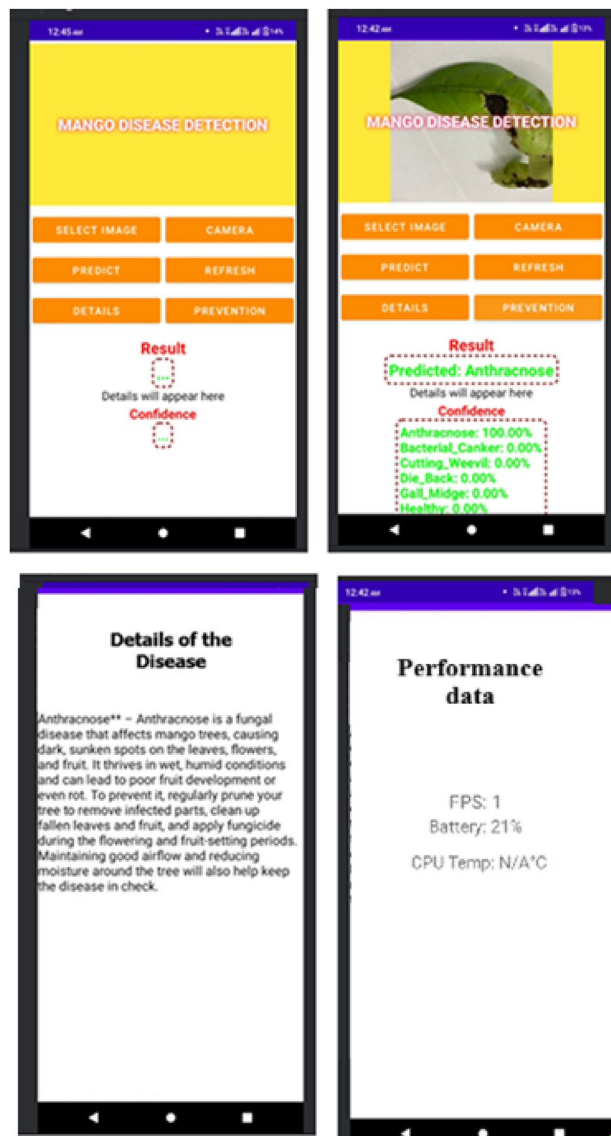


Fig. 23. Android App interface and sample result.



Fig. 24. Comparison of parameters of SOTA CNNs and mangoNet.

Data availability

The study was conducted using the primary dataset collected from the mango gardens in Rajshahi, Bangladesh. The Corresponding and Principal author is the owner of the data. The data will be made available on request. However, the study also utilized publicly available data, the link is [<https://www.kaggle.com/datasets/aryashah2k/mango-leaf-disease-dataset>].

Received: 3 October 2025; Accepted: 16 February 2026

Published online: 27 February 2026

References

- Rubab, N., Shan, M., & Hussain, A. IoT-based system for prediction of fungal disease attack on mango leaves. *Journal of Computing & Biomedical Informatics*, **8**(02), 963. <https://doi.org/10.56979/802/2025> (2025).
- Kiran, R., & Farhan, M. Climate variability and sustainable management of mango diseases: Emphasizing modern tools and forecasting models. In S. A. H. Naqvi, S. Ahmad, & M. Ahmed (Eds.), *Climate change and mango production: Potential adaptation and mitigation options* (pp. 379–413). Springer. https://doi.org/10.1007/978-3-031-87035-4_17 (2025).
- Pacal, I. et al. A systematic review of deep learning techniques for plant diseases. *Artif. Intell. Rev.* **57**(11), 304 (2024).
- Seetha, J. et al. Mango leaf disease classification using hybrid Coyote-Grey Wolf optimization tuned neural network model. *Multimedia Tools Appl.* **83**(6), 17699–17725 (2024).
- Thakur, P. S., Sheorey, T. & Ojha, A. VGG-ICNN: A lightweight CNN model for crop disease identification. *Multimedia Tools Appl.* **82**(1), 497–520 (2023).
- Fu, Y., Guo, L., & Huang, F. A lightweight CNN model for pepper leaf disease recognition in a human palm background. *Heliyon*, **10**(12), e33447. <https://doi.org/10.1016/j.heliyon.2024.e33447> (2024).
- Wang, A., Chen, H., Lin, Z., Han, J. & Ding, G. Repvit: Revisiting mobile cnn from vit perspective. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 15909–15920). (2024).
- Rakib, A. F., Rahman, R., Razi, A. A. & Hasan, A. T. A lightweight quantized CNN model for plant disease recognition. *Arab. J. Sci. Eng.* **49**(3), 4097–4108 (2024).
- Feng, W., Song, Q., Sun, G. & Zhang, X. Lightweight isotropic convolutional neural network for plant disease identification. *Agronomy* **13** (7), 1849 (2023).
- Alnamoly, M. H., Hady, A. A., Abd El-Kader, S. M., & El-Henawy, I. FL-ToLeD: An improved lightweight attention convolutional neural network model for tomato leaf diseases classification for low-end devices. *IEEE Access*, **12**, 73561–73580. <https://doi.org/10.1109/ACCESS.2024.3401733> (2024).
- Chakrabarty, A., Ahmed, S. T., Islam, M. F. U., Aziz, S. M. & Maidin, S. S. An interpretable fusion model integrating lightweight CNN and transformer architectures for rice leaf disease identification. *Ecol. Inform.* **82**, 102718 (2024).
- Lachure, J. & Doriya, R. Designing of lightweight deep learning framework for plant disease detection. *SN Comput. Sci.* **5**(6), 761 (2024).
- Hari, P. & Singh, M. P. A lightweight convolutional neural network for disease detection of fruit leaves. *Neural Comput. Appl.* **35** (20), 14855–14866 (2023).
- Prabu, M. & Chelliah, B. J. Mango leaf disease identification and classification using a CNN architecture optimized by crossover-based levy flight distribution algorithm. *Neural Comput. Appl.* **34**(9), 7311–7324 (2022).

15. Singh, U. P., Chouhan, S. S., Jain, S. & Jain, S. Multilayer convolution neural network for the classification of mango leaves infected by anthracnose disease. *IEEE Access* **7**, 43721–43729 (2019).
16. Saleem, R., Yuan, B., Kurugollu, F., Anjum, A. & Liu, L. Explaining deep neural networks: A survey on the global interpretation methods. *Neurocomputing* **513**, 165–180 (2022).
17. Alicioglu, G. & Sun, B. A survey of visual analytics for explainable artificial intelligence methods. *Computers Graphics*. **102**, 502–520 (2022).
18. Saleem, R., Shah, J. H., Sharif, M., & Ansari, G. J. Mango leaf disease identification using fully resolution convolutional network. *Computers, Materials & Continua*, **69**(3), 3581–3601. <https://doi.org/10.32604/cmc.2021.017700> (2021).
19. Wang, D. et al. A review of deep learning in multiscale agricultural sensing. *Remote Sens.* **14**(3), 559 (2022).
20. Manoharan, N., Thomas, V. J. & Dhas, D. A. S. Identification of mango leaf disease using deep learning. In *2021 Asian Conference on Innovation in Technology (ASIANCON)* (pp. 1–8). IEEE. (2021), August.
21. Ding, W., Abdel-Basset, M., Alrashdi, I. & Hawash, H. Next generation of computer vision for plant disease monitoring in precision agriculture: A contemporary survey, taxonomy, experiments, and future direction. *Inf. Sci.* **665**, 120338 (2024).
22. Mhamed, M., & Zhang, Z. AI technologies for apple leaf diseases identification: Scientometric analysis. In Z. Zhang, D. Zhu, C. Zuo, B. Han, P. Liu, & Z. Wang (Eds.), *Apple production technologies: From laboratory to practical applications: Cutting-edge innovative techniques* (pp. 137–161). Springer Nature Singapore. https://doi.org/10.1007/978-981-96-5747-6_7 (2025).
23. Mhamed, M. & Zhang, Z. Applications of the Internet of Things on Agriculture: Review and Future Apple Fruit Comprehensive Automation System. *Apple Production Technologies: From Laboratory to Practical Applications: Cutting-Edge Innovative Techniques*, 45–68. (2025).
24. Shaik, T. & Swamykan, S. I. Identification of diseases affecting mango leaves using deep learning models. In *International Conference on Artificial Intelligence: Towards Sustainable Intelligence* (pp. 132–144). Springer Nature Switzerland. (2023), September.
25. Thaseentaj, S., & Ilango, S. S. Deep convolutional neural networks for South Indian mango leaf disease detection and classification. *Computers, Materials & Continua*, **77**(3), 3593–3618. <https://doi.org/10.32604/cmc.2023.042496> (2023).
26. Shaheen, M., Abdelrahman, R. & Sarhan, A. M. Deep Learning Techniques for Detecting Lemon Leaf Diseases for Supporting Food Security. *Innovative AI Technologies Driving Sustainable Farming: Strategies for Improving Food Security*, 351. (2026).
27. Hossain, M. A., Sakib, S., Abdullah, H. M., & Arman, S. E. Deep learning for mango leaf disease identification: A vision transformer perspective. *Heliyon*, **10**(17), e36361. <https://doi.org/10.1016/j.heliyon.2024.e36361> (2024).
28. Hernández, I., Gutiérrez, S., Barrio, I., Iñiguez, R. & Tardaguila, J. In-field disease symptom detection and localisation using explainable deep learning: Use case for downy mildew in grapevine. *Comput. Electron. Agric.* **226**, 109478 (2024).
29. Fahim-Ul-Islam, M., Chakrabarty, A., Rahman, R., Moon, H. & Piran, M. J. Advancing mango leaf variant identification with a robust multi-layer perceptron model. *Sci. Rep.* **14**(1), 27406 (2024).
30. Sultana, S., Tasir, M. A. M., Nobel, S. N., Kabir, M. M. & Mridha, M. F. XAI-FruitNet: An explainable deep model for accurate fruit classification. *J. Agric. Food Res.* **18**, 101474 (2024).
31. Ramanarayanan, V. et al. Analysis of speech production real-time MRI. *Comput. Speech Lang.* **52**, 1–22 (2018).
32. Kunduracioglu, I. & Pacal, I. Advancements in deep learning for accurate classification of grape leaves and diagnosis of grape diseases. *J. Plant Dis. Prot.* **131** (3), 1061–1080 (2024).
33. Chen, Y. C., Wang, J. C., Lee, M. H., Liu, A. C. & Jiang, J. A. Enhanced detection of mango leaf diseases in field environments using MSMP-CNN and transfer learning. *Comput. Electron. Agric.* **227**, 109636 (2024).
34. Bezabh, Y. A. et al. Classification of mango disease using ensemble convolutional neural network. *Smart Agric. Technol.* **8**, 100476 (2024).
35. Salamai, A. A. Enhancing mango disease diagnosis through eco-informatics: A deep learning approach. *Ecol. Inform.* **77**, 102216 (2023).
36. Zhang, H., Sindagi, V. & Patel, V. M. Joint transmission map estimation and dehazing using deep networks. *IEEE Trans. Circuits Syst. Video Technol.* **30** (7), 1975–1986 (2019).
37. Cho, J., Kim, S., Min, D. & Sohn, K. Single image deraining using time-lapse data. *IEEE Trans. Image Process.* **29**, 7274–7289 (2020).
38. Kim, C. Y., Um, K. S. & Heo, S. W. A novel MobileNet with selective depth multiplier to compromise complexity and accuracy. *ETRI J* **45**(4), 666–677 (2023).
39. Zhao, L., Wang, L., Jia, Y. & Cui, Y. A lightweight deep neural network with higher accuracy. *PLoS One* **17**(8), e0271225 (2022).
40. Apicella, A., Isgrò, F. & Prevete, R. Don't push the button! Exploring data leakage risks in machine learning and transfer learning. *Artif. Intell. Rev.* **58**(11), 339 (2025).
41. Zheng, C., & Sun, D.-W. (2008). Image segmentation techniques. In D.-W. Sun (Ed.), *Computer vision technology for food quality evaluation* (pp. 37–56). Academic Press.
42. Steyerberg, E. W. et al. Poor performance of clinical prediction models: The harm of commonly applied methods. *J. Clin. Epidemiol.* **98**, 133–143 (2018).
43. Nguyen, H. T. et al. An improved MobileNet for disease detection on tomato leaves. *Adv. Technol. Innov.* **8** (3), 192–209 (2023).
44. Varma, T., Mate, P., Azeem, N. A., Sharma, S., & Singh, B. Automatic mango leaf disease detection using different transfer learning models. *Multimedia Tools and Applications*, **84**(11), 9185–9218. <https://doi.org/10.1007/s11042-024-19265-x> (2024).
45. Ahmed, J., Ahmed, F., Kabir, M. A., Ahad, M. T., Jadoon, M. A., Rehman, A. U., & Bermak, A. LBNet: An optimized lightweight CNN for mammographic breast cancer classification with XAI-based interpretability. *Scientific Reports*, **16**, 2013. <https://doi.org/10.1038/s41598-025-31642-6> (2025).
46. Ayon, R., Ahad, M. T., Song, B. & Li, Y. R-Net: A reliable and resource-efficient CNN for colorectal cancer detection with XAI integration. *arXiv preprint arXiv:2509.16251*. (2025).
47. Sagor, S., Ahad, M. T., Ahmed, F., Ayon, R. & Parvin, S. A study on deep convolutional neural networks, transfer learning, and Mnet model for cervical cancer detection. *arXiv preprint arXiv:2509.16250*. (2025).
48. Ahmed, M. E., Tuhin, H. H., Kayum, M. A., Islam, M. J. & Ahad, M. T. A vision transformer approach to potato leaf disease detection. In *2025 International Conference on Quantum Photonics, Artificial Intelligence*. (2025).
49. Mustofa, S., Emon, Y. R., Mamun, S. B., Akhy, S. A. & Ahad, M. T. A novel AI-driven model for student dropout risk analysis with explainable AI insights. *Computers and Education: Artificial Intelligence* **8**, 100352 (2025).
50. Mamun, S. B. et al. Grape guard: A YOLO-based mobile application for detecting grape leaf diseases. *J. Electron. Sci. Technol.* **23**(1), 100300 (2025).
51. Islam, R., Ahad, M. T., Ahmed, F., Song, B., & Li, Y. Mental health diagnosis from voice data using convolutional neural networks and vision transformers. *Journal of Voice*, **38**(6), 891–902. <https://doi.org/10.1016/j.jvoice.2024.10.010> (2024).
52. Khanom, S. et al. Detection of leg tremors in Parkinson's disease patients: An experimental wearable leg band solution. In *21st International Conference on Electrical Engineering, Computing*. (2024).
53. Ahad, M. T., Payel, I. J., Song, B. & Li, Y. DVS: Blood cancer detection using novel CNN-based ensemble approach. *arXiv preprint arXiv:2410.05272*. (2024).
54. Preanto, S. A., Ahad, M. T., Emon, Y. R., Mustofa, S. & Alamin, M. A study on deep feature extraction to detect and classify Acute Lymphoblastic Leukemia (ALL). *arXiv preprint arXiv:2409.06687*. (2024).
55. Ahad, M. T., Mustofa, S., Ahmed, F., Emon, Y. R. & Anu, A. D. A study on deep convolutional neural networks, transfer learning, and ensemble model for breast cancer detection. *arXiv preprint arXiv:2409.06699*. (2024).
56. Ahad, M. T., Mamun, S. B., Mustofa, S., Song, B. & Li, Y. A comprehensive study on blood cancer detection and classification using convolutional neural networks. *arXiv preprint arXiv:2409.06689*. (2024).

57. Preanto, S. A., Ahad, M. T., Emon, Y. R., Mustofa, S. & Alamin, M. A study on deep feature extraction to detect and classify Acute Lymphoblastic Leukemia (ALL). *arXiv e-prints*, *arXiv: 2409.06687*. (2024).
58. Preanto, S. A., Ahad, M. T., Emon, Y. R., Mustofa, S. & Alamin, M. A semantic segmentation approach on sweet orange leaf diseases detection utilizing YOLO. *arXiv e-prints*, *arXiv: 2409.06671*. (2024).
59. Ahad, M. T., Mustofa, S., Ahmed, F., Emon, Y. R. & Anu, A. D. A study on deep convolutional neural networks, transfer learning, and ensemble model for breast cancer detection. *arXiv e-prints*, *arXiv: 2409.06699*. (2024).
60. Bhowmik, A. C. et al. A customized vision transformer for accurate detection and classification of Java plum leaf disease. *Smart Agric. Technol.* **8**, 100500 (2024).
61. Emon, Y. R., Ahad, M. T. & Rabbany, G. Multi-format open-source sweet orange leaf dataset for disease detection, classification, and analysis. *Data Brief*. **55**, 110713 (2024).
62. Mamun, S. B., Ahad, M. T., Morshed, M. M., Hossain, N. & Emon, Y. R. Scratch vision transformer model for diagnosis grape leaf disease. In *Proceedings of the Fifth International Conference on Trends in Computational*. (2024).
63. Ahmed, F., Emon, Y. R., Ahad, M. T., Munna, M. H. & Mamun, S. B. A fuzzy-based vision transformer model for tea leaf disease detection. In *Proceedings of the Fifth International Conference on Trends in Computational*. (2024).
64. Bhowmik, A. C., Ahad, M. T., Emon, Y. R., Ahmed, F., & Song, B. A customized vision transformer for accurate detection and classification of Java plum leaf disease. *Smart Agricultural Technology*, **8**, 100500. <https://doi.org/10.1016/j.atech.2024.100500> (2024).
65. Ahad, M. T., Mustofa, S., Sarker, A. & Emon, Y. R. Bdpapayaleaf: A dataset of papaya leaf for disease detection, classification, and analysis. *Classification, and Analysis*. (2024).
66. Ahad, M. T., Song, B. & Li, Y. A comparison of convolutional neural network, transfer learning, and ensemble technique for brain tumour detection or classification. *Transfer Learning and Ensemble Technique for Brain Tumour Detection*. (2024).
67. Bhowmik, A. C., Ahad, M. T. & Emon, Y. R. Machine learning-based Jamun leaf disease detection: A comprehensive review. *arXiv preprint arXiv:2311.15741*. (2023).
68. Mamun, S. B., Ahad, M. T., Morshed, M. M., Hossain, N. & Emon, Y. R. Scratch vision transformer model for diagnosis grape leaf disease. In *International Conference on Trends in Computational and Cognitive*. (2023).
69. Ahmed, F., Emon, Y. R., Ahad, M. T., Munna, M. H. & Mamun, S. B. A fuzzy-based vision transformer model for tea leaf disease detection. In *International Conference on Trends in Computational and Cognitive*. (2023).
70. Biplob, T. I., Rabbany, G., Emon, Y. R., Ahad, M. T. & Fimu, F. A. An optimized vision-based transformer for lung cancer detection. In *International Conference on Trends in Computational and Cognitive*. (2023).
71. Ahmed, F., Ahad, M. T., & Emon, Y. R. Machine learning-based tea leaf disease detection: A comprehensive review. *arXiv*. <https://arxiv.org/abs/2311.03240> (2023).
72. Ahad, M. T., Li, Y., Song, B. & Bhuiyan, T. Comparison of CNN-based deep learning architectures for rice diseases classification. *Artif. Intell. Agric.* **9**, 22–35 (2023).

Acknowledgements

We thank the mango orchard owners who allowed us to collect the mango leaf image.

Author contributions

Md Taimur Ahad: Writing – original draft, project administration, methodology, investigation, formal analysis, conceptualization.

Funding statement

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declarations

Competing interests

The authors declare no competing interests.

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